

# Pathway Analysis Report

This report contains the pathway analysis results for the submitted sample ". Analysis was performed against Reactome version 86 on 01/10/2023. The web link to these results is:

<https://reactome.org/PathwayBrowser/#/ANALYSIS=MjAyMzEwMDExNTQ1NDBfMjIwNDY%3D>

Please keep in mind that analysis results are temporarily stored on our server. The storage period depends on usage of the service but is at least 7 days. As a result, please note that this URL is only valid for a limited time period and it might have expired.

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# 1. Introduction

Reactome is a curated database of pathways and reactions in human biology. Reactions can be considered as pathway 'steps'. Reactome defines a 'reaction' as any event in biology that changes the state of a biological molecule. Binding, activation, translocation, degradation and classical biochemical events involving a catalyst are all reactions. Information in the database is authored by expert biologists, entered and maintained by Reactome's team of curators and editorial staff. Reactome content frequently cross-references other resources e.g. NCBI, Ensembl, UniProt, KEGG (Gene and Compound), ChEBI, PubMed and GO. Orthologous reactions inferred from annotation for Homo sapiens are available for 14 non-human species including mouse, rat, chicken, puffer fish, worm, fly and yeast. Pathways are represented by simple diagrams following an SBGN-like format.

Reactome's annotated data describe reactions possible if all annotated proteins and small molecules were present and active simultaneously in a cell. By overlaying an experimental dataset on these annotations, a user can perform a pathway over-representation analysis. By overlaying quantitative expression data or time series, a user can visualize the extent of change in affected pathways and its progression. A binomial test is used to calculate the probability shown for each result, and the p-values are corrected for the multiple testing (Benjamini-Hochberg procedure) that arises from evaluating the submitted list of identifiers against every pathway.

To learn more about our Pathway Analysis, please have a look at our relevant publications:

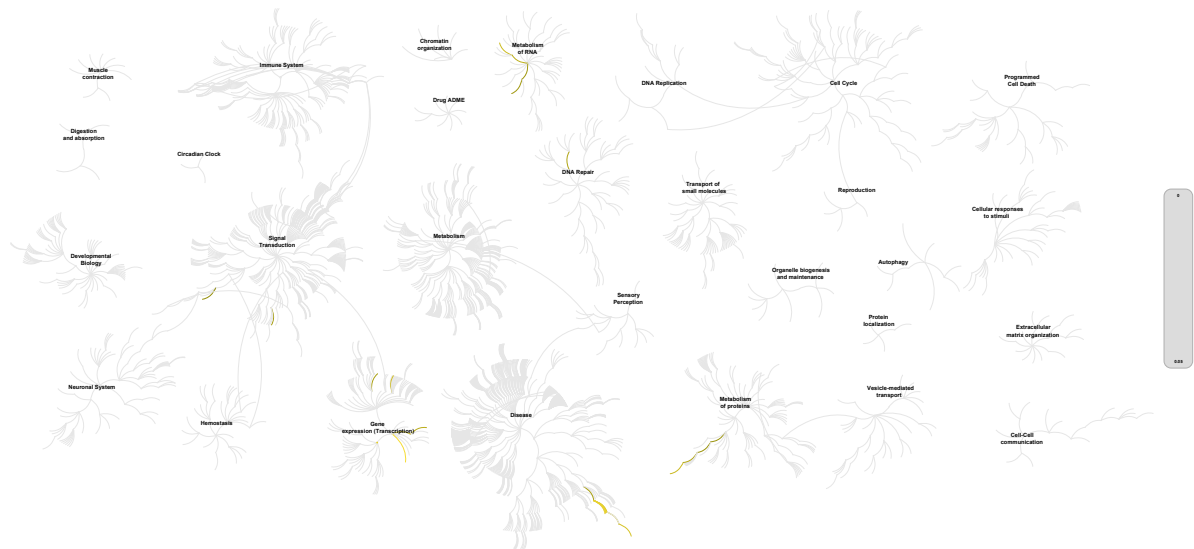
Fabregat A, Sidiropoulos K, Garapati P, Gillespie M, Hausmann K, Haw R, ... D'Eustachio P (2016). The reactome pathway knowledgebase. *Nucleic Acids Research*, 44(D1), D481–D487. <https://doi.org/10.1093/nar/gkv1351>. 

Fabregat A, Sidiropoulos K, Viteri G, Forner O, Marin-Garcia P, Arnau V, ... Hermjakob H (2017). Reactome pathway analysis: a high-performance in-memory approach. *BMC Bioinformatics*, 18. 

## 2. Properties

- This is an **overrepresentation** analysis: A statistical (hypergeometric distribution) test that determines whether certain Reactome pathways are over-represented (enriched) in the submitted data. It answers the question 'Does my list contain more proteins for pathway X than would be expected by chance?' This test produces a probability score, which is corrected for false discovery rate using the Benjamini-Hochberg method. [↗](#)
- 2 out of 2 identifiers in the sample were found in Reactome, where 65 pathways were hit by at least one of them.
- All non-human identifiers have been converted to their human equivalent. [↗](#)
- IntAct interactors were included to increase the analysis background. This greatly increases the size of Reactome pathways, which maximises the chances of matching your submitted identifiers to the expanded pathway, but will include interactors that have not undergone manual curation by Reactome and may include interactors that have no biological significance, or unexplained relevance.
- This report is filtered to show only results for species 'Homo sapiens' and resource 'UniProt'.
- The unique ID for this analysis (token) is MjAyMzEwMDExNTQ1NDBfMjIwNDY%3D. This ID is valid for at least 7 days in Reactome's server. Use it to access Reactome services with your data.

### 3. Genome-wide overview



This figure shows a genome-wide overview of the results of your pathway analysis. Reactome pathways are arranged in a hierarchy. The center of each of the circular "bursts" is the root of one top-level pathway, for example "DNA Repair". Each step away from the center represents the next level lower in the pathway hierarchy. The color code denotes over-representation of that pathway in your input dataset. Light grey signifies pathways which are not significantly over-represented.

## 4. Most significant pathways

The following table shows the 25 most relevant pathways sorted by p-value.

| Pathway name                                                                                           | Entities  |       |          |       | Reactions  |          |
|--------------------------------------------------------------------------------------------------------|-----------|-------|----------|-------|------------|----------|
|                                                                                                        | found     | ratio | p-value  | FDR*  | found      | ratio    |
| RNA polymerase II transcribes snRNA genes                                                              | 2 / 220   | 0.012 | 7.98e-04 | 0.052 | 5 / 11     | 7.60e-04 |
| Tat-mediated HIV elongation arrest and recovery                                                        | 1 / 32    | 0.002 | 0.007    | 0.084 | 3 / 3      | 2.07e-04 |
| Pausing and recovery of Tat-mediated HIV elongation                                                    | 1 / 32    | 0.002 | 0.007    | 0.084 | 2 / 2      | 1.38e-04 |
| HIV elongation arrest and recovery                                                                     | 1 / 33    | 0.002 | 0.007    | 0.084 | 3 / 3      | 2.07e-04 |
| Pausing and recovery of HIV elongation                                                                 | 1 / 33    | 0.002 | 0.007    | 0.084 | 2 / 2      | 1.38e-04 |
| Formation of HIV elongation complex in the absence of HIV Tat                                          | 1 / 55    | 0.003 | 0.012    | 0.084 | 1 / 2      | 1.38e-04 |
| Gene expression (Transcription)                                                                        | 4 / 6,234 | 0.329 | 0.012    | 0.084 | 22 / 1,090 | 0.075    |
| Tat-mediated elongation of the HIV-1 transcript                                                        | 1 / 76    | 0.004 | 0.016    | 0.084 | 4 / 8      | 5.52e-04 |
| Formation of HIV-1 elongation complex containing HIV-1 Tat                                             | 1 / 76    | 0.004 | 0.016    | 0.084 | 1 / 5      | 3.45e-04 |
| HIV Transcription Elongation                                                                           | 1 / 81    | 0.004 | 0.017    | 0.084 | 4 / 15     | 0.001    |
| RUNX1 interacts with co-factors whose precise effect on RUNX1 targets is not known                     | 1 / 97    | 0.005 | 0.02     | 0.084 | 1 / 5      | 3.45e-04 |
| Deadenylation of mRNA                                                                                  | 1 / 99    | 0.005 | 0.021    | 0.084 | 2 / 4      | 2.76e-04 |
| Translation initiation complex formation                                                               | 1 / 100   | 0.005 | 0.021    | 0.084 | 1 / 2      | 1.38e-04 |
| Deadenylation-dependent mRNA decay                                                                     | 1 / 133   | 0.007 | 0.028    | 0.084 | 2 / 10     | 6.91e-04 |
| Formation of RNA Pol II elongation complex                                                             | 1 / 140   | 0.007 | 0.029    | 0.084 | 2 / 2      | 1.38e-04 |
| RNA Polymerase II Transcription Elongation                                                             | 1 / 145   | 0.008 | 0.03     | 0.084 | 5 / 8      | 5.52e-04 |
| Activation of the mRNA upon binding of the cap-binding complex and eIFs, and subsequent binding to 43S | 1 / 151   | 0.008 | 0.032    | 0.084 | 1 / 6      | 4.14e-04 |
| Transcription-Coupled Nucleotide Excision Repair (TC-NER)                                              | 1 / 164   | 0.009 | 0.034    | 0.084 | 1 / 17     | 0.001    |
| Transcription of the HIV genome                                                                        | 1 / 176   | 0.009 | 0.037    | 0.084 | 15 / 47    | 0.003    |
| TP53 Regulates Transcription of DNA Repair Genes                                                       | 1 / 177   | 0.009 | 0.037    | 0.084 | 3 / 17     | 0.001    |
| Regulation of PTEN gene transcription                                                                  | 1 / 183   | 0.01  | 0.038    | 0.084 | 2 / 15     | 0.001    |

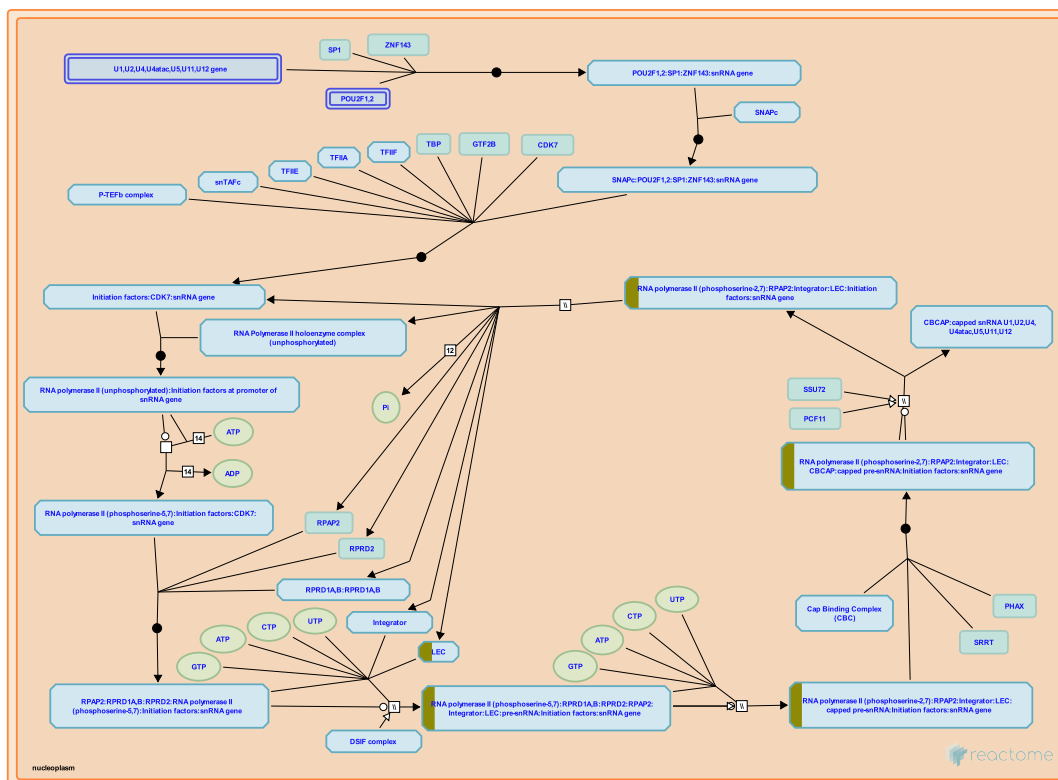
| Pathway name                                                              | Entities |       |         |       | Reactions |          |
|---------------------------------------------------------------------------|----------|-------|---------|-------|-----------|----------|
|                                                                           | found    | ratio | p-value | FDR*  | found     | ratio    |
| Nonsense Mediated Decay (NMD) enhanced by the Exon Junction Complex (EJC) | 1 / 201  | 0.011 | 0.042   | 0.084 | 1 / 5     | 3.45e-04 |
| Nonsense-Mediated Decay (NMD)                                             | 1 / 201  | 0.011 | 0.042   | 0.084 | 1 / 6     | 4.14e-04 |
| Cap-dependent Translation Initiation                                      | 1 / 212  | 0.011 | 0.044   | 0.088 | 1 / 18    | 0.001    |
| Eukaryotic Translation Initiation                                         | 1 / 215  | 0.011 | 0.045   | 0.089 | 1 / 21    | 0.001    |

\* False Discovery Rate

## 5. Pathways details

For every pathway of the most significant pathways, we present its diagram, as well as a short summary, its bibliography and the list of inputs found in it.

1. RNA polymerase II transcribes snRNA genes ([R-HSA-6807505](#))



**Cellular compartments:** nucleoplasm.

Small nuclear RNAs (snRNAs) play key roles in splicing and some of them, specifically the U1 and U2 snRNAs, are encoded by multicopy snRNA gene clusters containing tandem arrays of genes, about 30 in the RNU1 cluster (Bernstein et al. 1985) and about 10-20 in the RNU2 cluster (Van Ardsell and Weiner 1984). Whereas U6 snRNA genes are transcribed by RNA polymerase III, U1, U2, U4, U4atac, U5, U11, and U12 genes are transcribed by RNA polymerase II. Transcription of the U1 and U2 genes has been most extensively studied and the other snRNA genes as well as other genes with similar promoter structures, for example the SNORD13 gene, are inferred to be transcribed by similar reactions. The snRNA genes transcribed by RNA polymerase II are distinguished from mRNA-encoding genes by the presence of a proximal sequence element (PSE) rather than a TATA box and the presence of the Integrator complex rather than the Mediator complex (reviewed in Egloff et al. 2008, Jawdeker and Henry 2008).

The snRNA genes are among the most rapidly transcribed genes in the genome. The 5' transcribed region of the U2 snRNA gene is largely single-stranded during interphase and metaphase (Pavelitz et al. 2008) and chromatin within the transcribed region is cleared of nucleosomes (O'Reilly et al. 2014). Transcriptional activation of the RNA polymerase II transcribed snRNA genes begins with binding of transcription factors to the distal sequence element (DSE) of the promoter (reviewed in Hernandez 2001, Egloff et al. 2008, Jawdeker and Henry 2008). The factors, which include POU2F1 (Oct-1), POU2F2 (Oct-2), ZNF143 (Staf) and Sp1, promote binding of the SNAPc complex (also known as PTF and PBP) to the PSE. SNAPc helps clear the gene of nucleosomes (O'Reilly et al. 2014) and recruits initiation factors (TFIIA, TFIIB, TFIIE, TFIIIF, and snTAFc:TBP) which recruit RNA polymerase II. Phosphorylation of the C-terminal domain (CTD) of RNA polymerase II (reviewed in Egloff and Murphy 2008) by CDK7 recruits RPAP2 and the Integrator complex, which is required for later processing of the 3' end of the pre-snRNA transcript (reviewed in Chen and Wagner 2010, Bailat and Wagner 2015). The Little Elongation Complex (LEC) also appears to bind around the time of transcription initiation (Hu et al. 2013). As transcription proceeds, RPAP2 dephosphorylates serine-5 and P-TEFb phosphorylates serine-2 of the CTD. As transcription reaches the end of the snRNA gene serine-7 of the CTD is phosphorylated. These marks serve to bind protein complexes and are required for 3' processing of the pre-snRNA (reviewed in Egloff and Murphy 2008). After transcription proceeds through the conserved 3' processing sequence of the pre-snRNA the Integrator complex cleaves the pre-snRNA. Transcription then terminates downstream in a less well characterized reaction that requires elements of the polyadenylation system.

## References

- Pavelitz T, Elco CP, Bailey AD & Weiner AM (2008). Human U2 snRNA genes exhibit a persistently open transcriptional state and promoter disassembly at metaphase. *Mol. Cell. Biol.*, 28, 3573-88. [↗](#)
- Laitem C, Murphy S, Dienstbier M, O'Reilly D, Zaborowska J & Kuznetsova OV (2014). Human snRNA genes use polyadenylation factors to promote efficient transcription termination. *Nucleic Acids Res.*, 42, 264-75. [↗](#)
- Eissenberg JC, Smith ER, Lin C, Hu D, Shilatifard A, Gao X, ... Garruss AS (2013). The little elongation complex functions at initiation and elongation phases of snRNA gene transcription. *Mol. Cell*, 51, 493-505. [↗](#)
- Murphy S, O'Reilly D & Egloff S (2008). Expression of human snRNA genes from beginning to end. *Biochem. Soc. Trans.*, 36, 590-4. [↗](#)
- Van Arsdel SW & Weiner AM (1984). Human genes for U2 small nuclear RNA are tandemly repeated. *Mol. Cell. Biol.*, 4, 492-9. [↗](#)

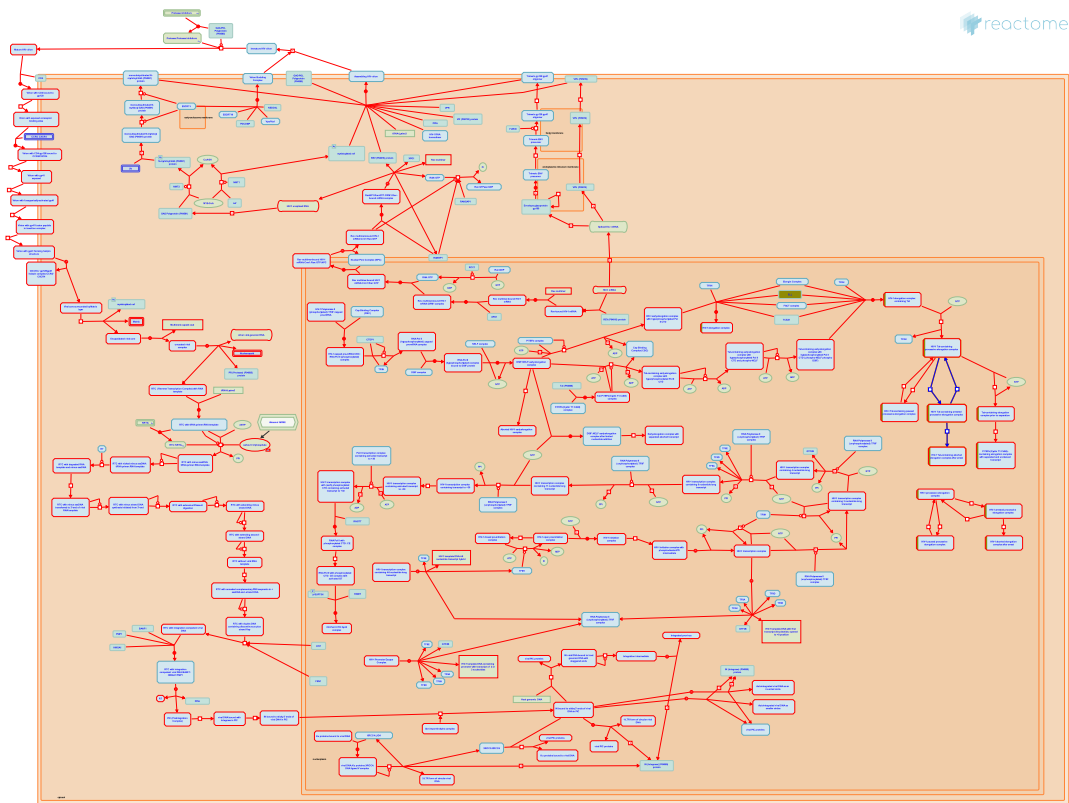
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| 2015-11-02 | Edited   | May B       |
| 2015-11-02 | Authored | May B       |
| 2015-11-02 | Created  | May B       |
| 2016-02-19 | Reviewed | Hernandez N |
| 2023-08-26 | Modified | Wright A    |

**1 submitted entities found in this pathway, mapping to 2 Reactome entities**

| Input | UniProt Id     |
|-------|----------------|
| ELL   | P55199, Q9HB65 |

2. Tat-mediated HIV elongation arrest and recovery (R-HSA-167243)



**Cellular compartments:** nucleoplasm.

**Diseases:** Human immunodeficiency virus infectious disease.

RNA Pol II arrest is believed to be a result of irreversible backsliding of the enzyme by ~7-14 nucleotides. TFIIS reactivates arrested RNA Pol II by promoting the excision of nascent transcript ~7-14 nucleotides upstream of the 3' end.

References

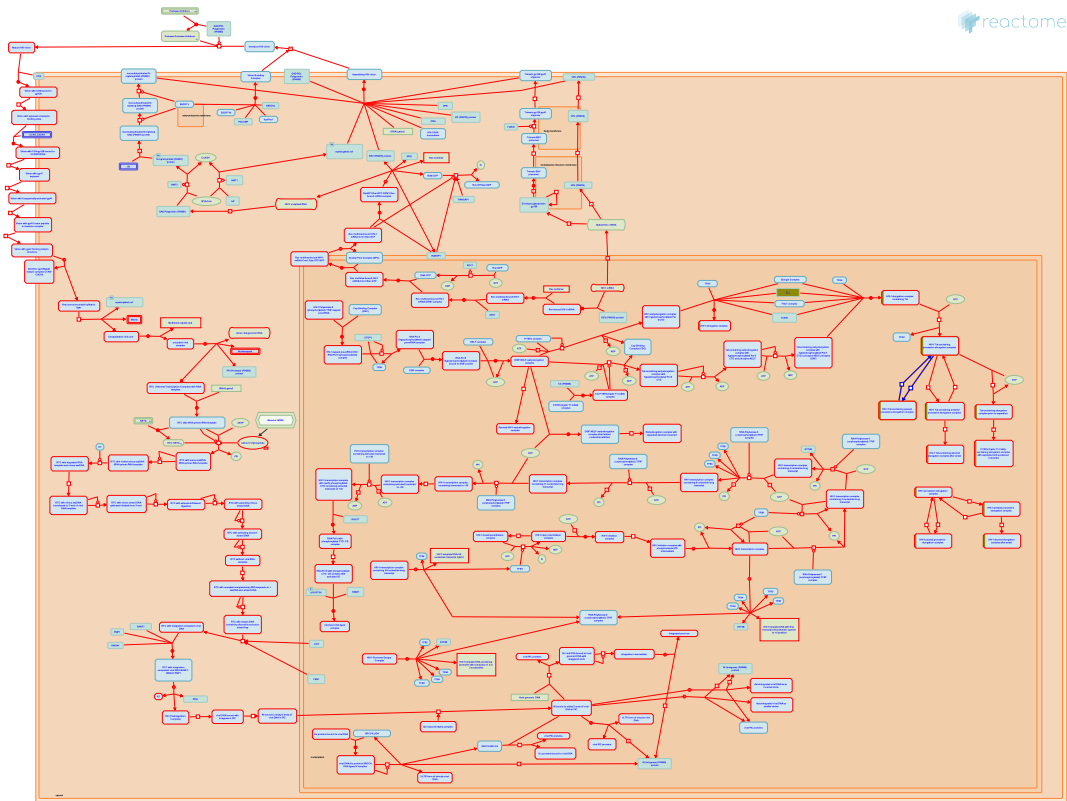
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| Date       | Action   | Author     |
|------------|----------|------------|
| 2005-10-14 | Created  | Matthews L |
| 2023-03-08 | Modified | Matthews L |

1 submitted entities found in this pathway, mapping to 1 Reactome entities

| Input | UniProt Id |
|-------|------------|
| ELL   | P55199     |

3. Pausing and recovery of Tat-mediated HIV elongation (R-HSA-167238)



**Cellular compartments:** nucleoplasm.

**Diseases:** Human immunodeficiency virus infectious disease.

After Pol II pauses by back tracking 2 -4 nucleotides on the HIV-1 template, elongation of the HIV-1 transcript resumes.

References

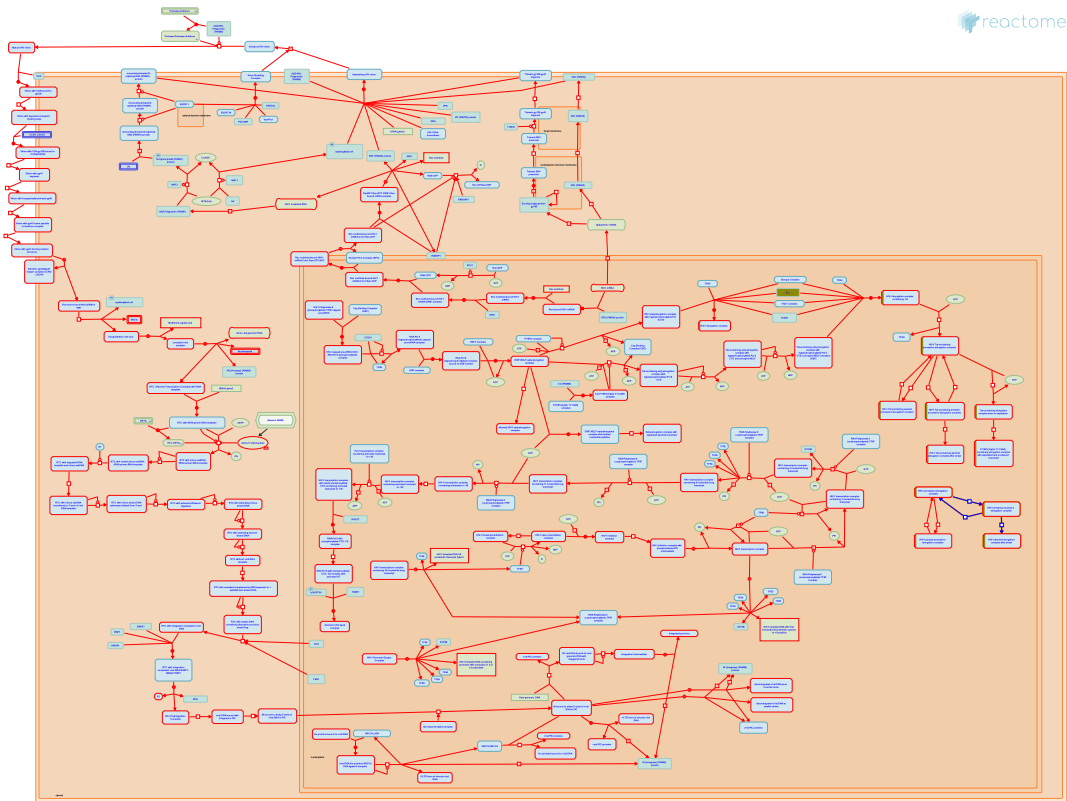
Edit history

| Date       | Action   | Author     |
|------------|----------|------------|
| 2005-10-14 | Created  | Matthews L |
| 2023-03-08 | Modified | Matthews L |

1 submitted entities found in this pathway, mapping to 1 Reactome entities

| Input | UniProt Id |
|-------|------------|
| ELL   | P55199     |

4. HIV elongation arrest and recovery (R-HSA-167287)



**Cellular compartments:** nucleoplasm.

**Diseases:** Human immunodeficiency virus infectious disease.

RNA Pol II arrest is believed to be a result of irreversible backsliding of the enzyme by ~7-14 nucleotides. TFIIS reactivates arrested RNA Pol II by promoting the excision of nascent transcript ~7-14 nucleotides upstream of the 3' end.

References

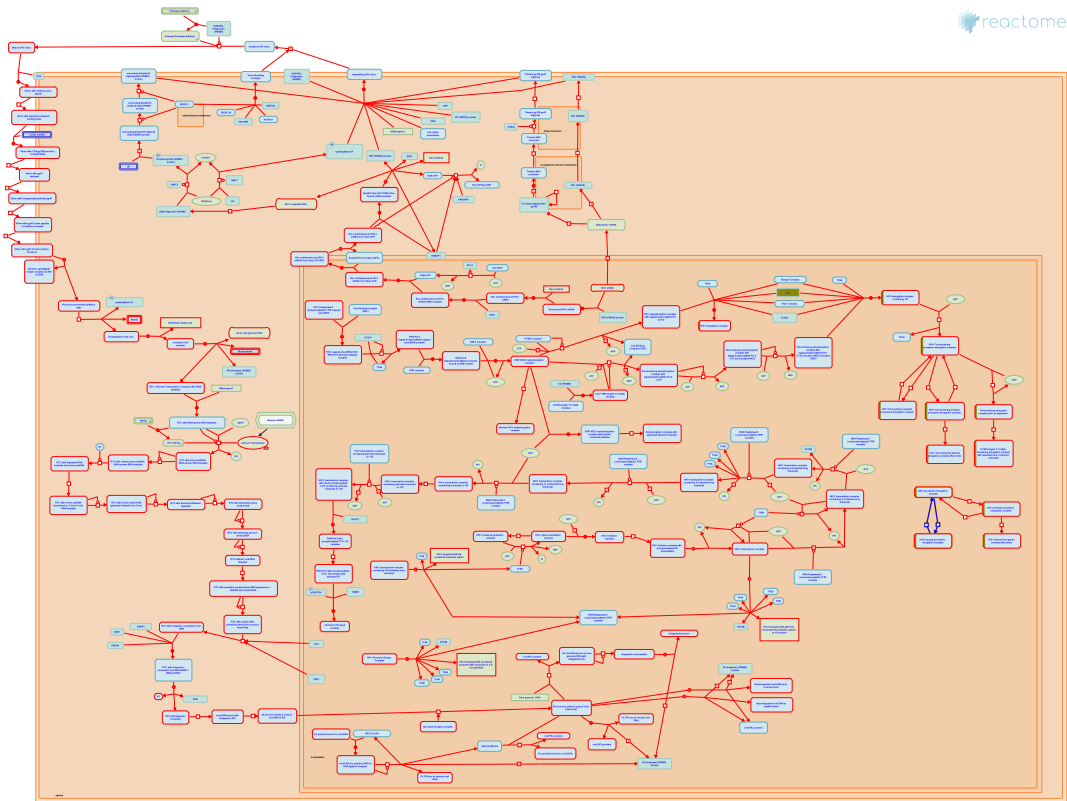
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|------------|----------|------------|
| 2005-10-14 | Created  | Matthews L |
| 2023-03-08 | Modified | Matthews L |

1 submitted entities found in this pathway, mapping to 1 Reactome entities

| Input | UniProt Id |
|-------|------------|
| ELL   | P55199     |

5. Pausing and recovery of HIV elongation (R-HSA-167290)



**Cellular compartments:** nucleoplasm.

**Diseases:** Human immunodeficiency virus infectious disease.

After Pol II pauses by back tracking 2 -4 nuleotides on the HIV-1 template, elongation of the HIV-1 transcript resumes.

References

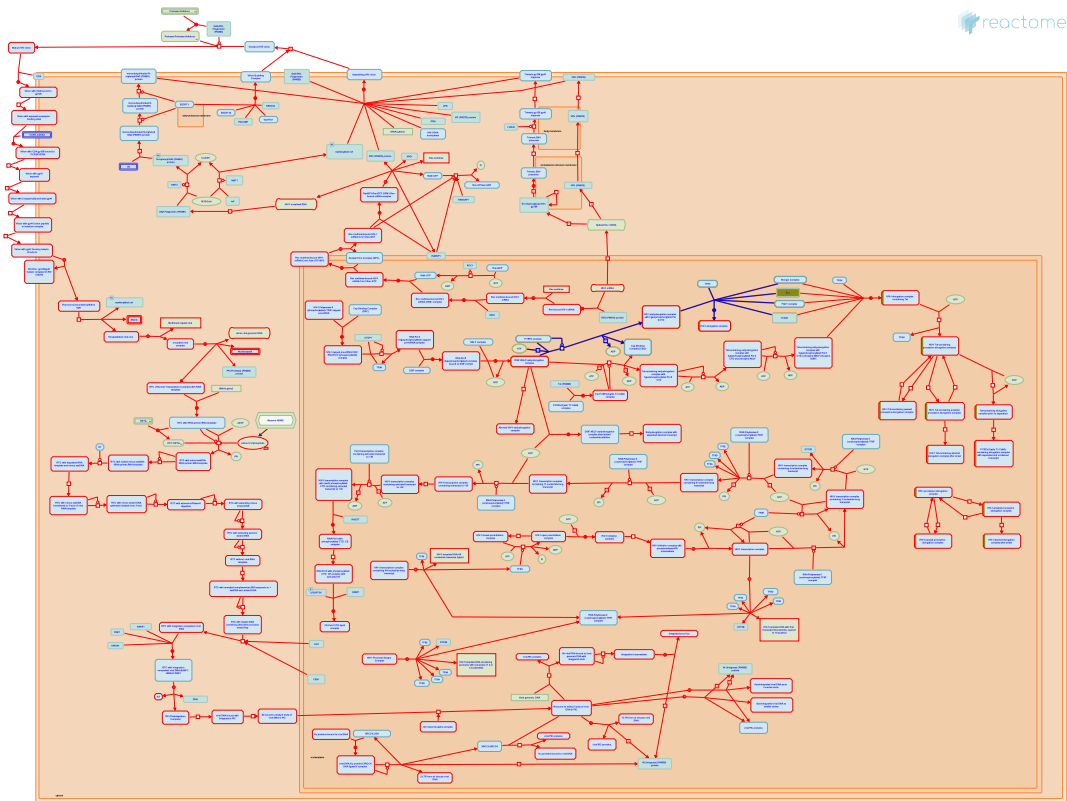
Edit history

| Date       | Action   | Author     |
|------------|----------|------------|
| 2005-10-14 | Created  | Matthews L |
| 2023-03-08 | Modified | Matthews L |

1 submitted entities found in this pathway, mapping to 1 Reactome entities

| Input | UniProt Id |
|-------|------------|
| ELL   | P55199     |

6. Formation of HIV elongation complex in the absence of HIV Tat (R-HSA-167152)



**Cellular compartments:** nucleoplasm.

**Diseases:** Human immunodeficiency virus infectious disease.

**Inferred from:** Formation of RNA Pol II elongation complex .

During the formation of the HIV elongation complex in the absence of HIV Tat, elongation factors are recruited to form the HIV-1 elongation complex (Hill and Sundquist 2013) and P-TEFb complex hyperphosphorylates RNA Pol II CTD (Hermann and Rice, 2005, Zhou et al., 2000).

References

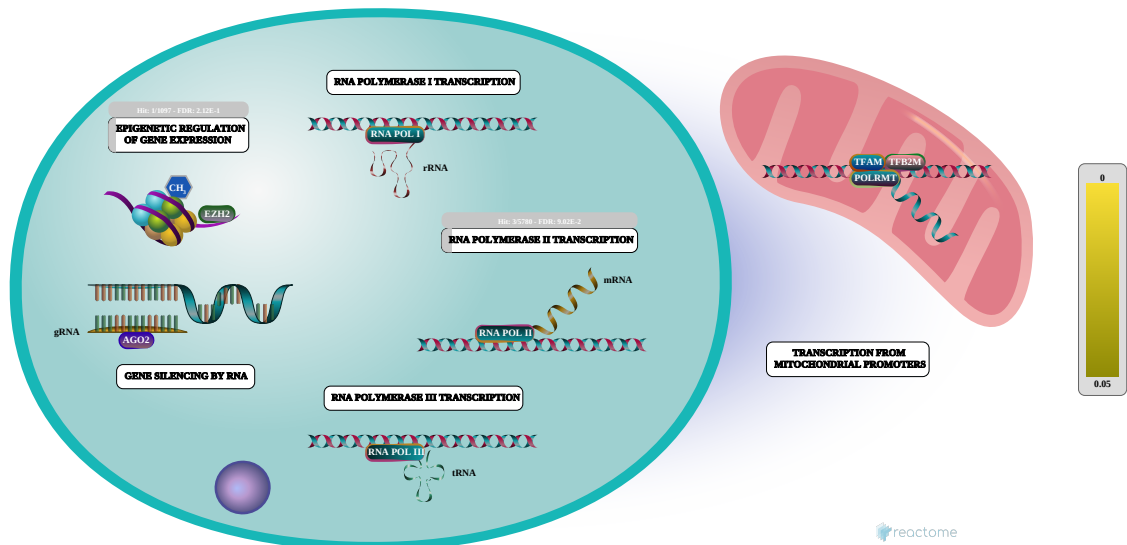
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|------------|----------|---------------------|
| 2005-07-27 | Edited   | Matthews L          |
| 2005-07-27 | Authored | Rice AP, Matthews L |
| 2005-10-11 | Created  | Matthews L          |
| 2023-03-08 | Modified | Matthews L          |

1 submitted entities found in this pathway, mapping to 1 Reactome entities

| Input | UniProt Id |
|-------|------------|
| ELL   | P55199     |

7. Gene expression (Transcription) (R-HSA-74160)



Gene expression encompasses transcription and translation and the regulation of these processes. RNA Polymerase I Transcription produces the large preribosomal RNA transcript (45S pre-rRNA) that is processed to yield 18S rRNA, 28S rRNA, and 5.8S rRNA, accounting for about half the RNA in a cell. RNA Polymerase II transcription produces messenger RNAs (mRNA) as well as a subset of non-coding RNAs including many small nucleolar RNAs (snRNA) and microRNAs (miRNA). RNA Polymerase III Transcription produces transfer RNAs (tRNA), 5S RNA, 7SL RNA, and U6 snRNA. Transcription from mitochondrial promoters is performed by the mitochondrial RNA polymerase, POLRMT, to yield long transcripts from each DNA strand that are processed to yield 12S rRNA, 16S rRNA, tRNAs, and a few RNAs encoding components of the electron transport chain. Regulation of gene expression can be divided into epigenetic regulation, transcriptional regulation, and post-transcription regulation (comprising translational efficiency and RNA stability). Epigenetic regulation of gene expression is the result of heritable chemical modifications to DNA and DNA-binding proteins such as histones. Epigenetic changes result in altered chromatin complexes that influence transcription. Gene Silencing by RNA mostly occurs post-transcriptionally but can also affect transcription. Small RNAs originating from the genome (miRNAs) or from exogenous RNA (siRNAs) are processed and transferred to the RNA-induced silencing complex (RISC), which interacts with complementary RNA to cause cleavage, translational inhibition, or transcriptional inhibition.

References

Edit history

| Date       | Action   | Author                                                             |
|------------|----------|--------------------------------------------------------------------|
| 2003-09-11 | Authored | Larsson NG, Gustafsson CM, Comai L, Reinberg D, Timmers HTM et al. |
| 2003-09-11 | Created  | Proudfoot NJ, Kornblihtt AR                                        |
| 2008-12-03 | Authored | Caudy M, Proudfoot NJ, Kornblihtt AR, D'Eustachio P                |
| 2016-12-29 | Revised  | D'Eustachio P                                                      |
| 2023-08-24 | Edited   | Joshi-Tope G                                                       |

| Date       | Action   | Author                    |
|------------|----------|---------------------------|
| 2023-08-24 | Reviewed | Paule M, Willis I, Zhao X |
| 2023-08-26 | Modified | Wright A                  |

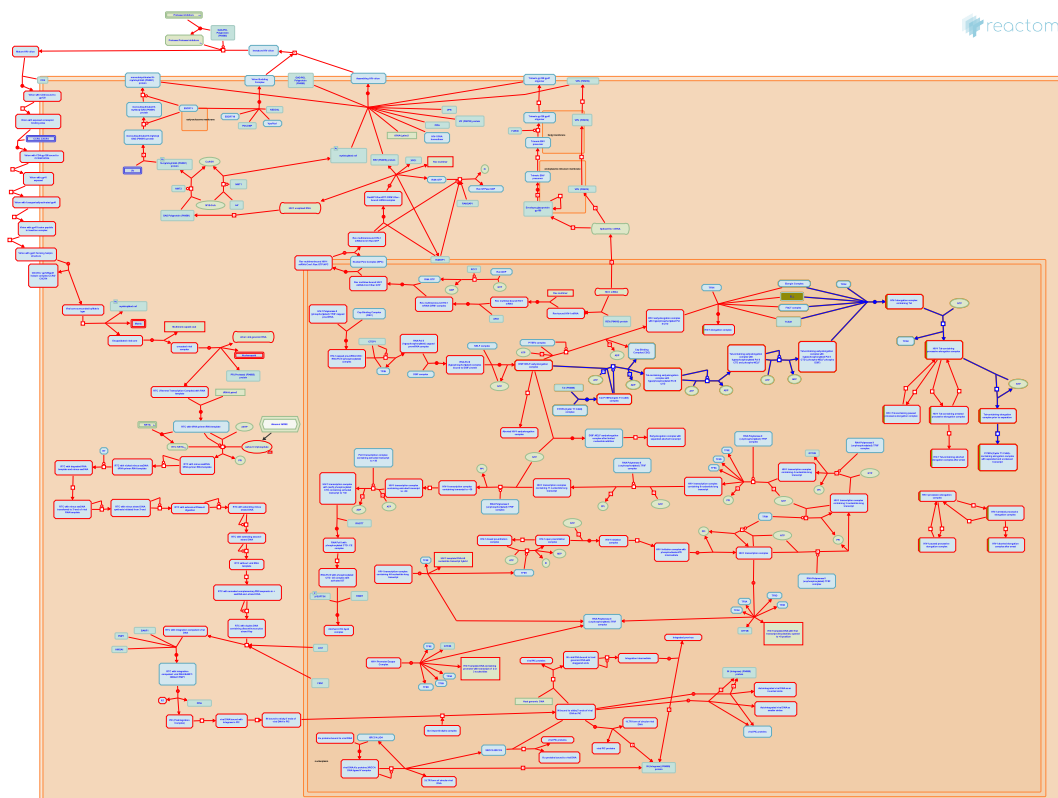
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| Input | UniProt Id | Input | UniProt Id     |
|-------|------------|-------|----------------|
| CBX6  | O95503     | ELL   | P55199, Q9HB65 |

**Interactors found in this pathway (2)**

| Input | UniProt Id | Interacts with | Input | UniProt Id | Interacts with |
|-------|------------|----------------|-------|------------|----------------|
| ELL   | P55199     | Q9UHB7, Q96JC9 | WDR37 | Q9Y2I8     | Q96ES7         |

## 8. Tat-mediated elongation of the HIV-1 transcript (R-HSA-167246)



**Cellular compartments:** nucleoplasm.

**Diseases:** Human immunodeficiency virus infectious disease.

The Tat protein is a viral transactivator protein that regulates HIV-1 gene expression by controlling RNA Pol II-mediated elongation (reviewed in Karn 1999; Taube et al. 1999; Liou et al. 2004; Barboric and Peterlin 2005). Tat appears to be required in order to overcome the arrest of RNA Pol II by the negative transcriptional elongation factors DSIF and NELF (Wada et al. 1998; Yamaguchi et al. 1999; Yamaguchi et al. 2002; Fujinaga et al. 2004). While Pol II can associate with the proviral LTR and initiate transcription in the absence of Tat, these polymerase complexes are non-processive and dissociate from the template prematurely producing very short transcripts (Kao et al. 1987). Tat associates with the RNA element, TAR, which forms a stem loop structure in the leader RNA sequence (Dingwall et al. 1989). Tat also associates with the cellular kinase complex P-TEFb(Cyclin T1:Cdk9) and recruits it to the TAR stem loop structure (Herrmann, 1995) (Wei et al. 1998). This association between Tat, TAR and P-TEFb(Cyclin T1:Cdk9) is believed to bring the catalytic subunit of this kinase complex (Cdk9) in close proximity to Pol II where it hyperphosphorylates the CTD of RNA Pol II (Zhou et al. 2000). The RD subunits of NELF and the SPT5 subunit of DSIF, which associate through RD with the bottom stem of TAR, are also phosphorylated by P-TEFb(Cyclin T1:Cdk9) (Yamaguchi et al. 2002; Fujinaga et al. 2004; Ivanov et al. 2000). Phosphorylation of RD results in its dissociation from TAR. Thus, Tat appears to facilitate transcriptional elongation of the HIV-1 transcript by hyperphosphorylating the RNA Pol II CTD and by removing the negative transcription elongation factors from TAR. In addition, there is evidence that the association of Tat with P-TEFb(Cyclin T1:Cdk9) alters the substrate specificity of P-TEFb enhancing phosphorylation of ser5 residues in the CTD of RNA Pol II (Zhou et al. 2000).

## References

Peterlin BM & Barboric M (2005). A new paradigm in eukaryotic biology: HIV Tat and the control of transcriptional elongation. PLoS Biol, 3, e76. [↗](#)

Skinner MA, Karn J, Green SM, Singh M, Ernberg I, Dingwall C, ... Valerio R (1989). Human immunodeficiency virus 1 tat protein binds trans-activation-responsive region (TAR) RNA in vitro. Proc Natl Acad Sci U S A, 86, 6925-9. [↗](#)

Peterlin BM, Irwin D, Taube R, Huang Y, Fujinaga K & Kurosu T (2004). Dynamics of human immunodeficiency virus transcription: P-TEFb phosphorylates RD and dissociates negative effectors from the transactivation response element. Mol Cell Biol, 24, 787-95. [↗](#)

Karn J (1999). Tackling Tat. J Mol Biol, 293, 235-54. [↗](#)

Gaynor RB, Kwak YT, Ivanov D & Guo J (2000). Domains in the SPT5 protein that modulate its transcriptional regulatory properties. Mol Cell Biol, 20, 2970-83. [↗](#)

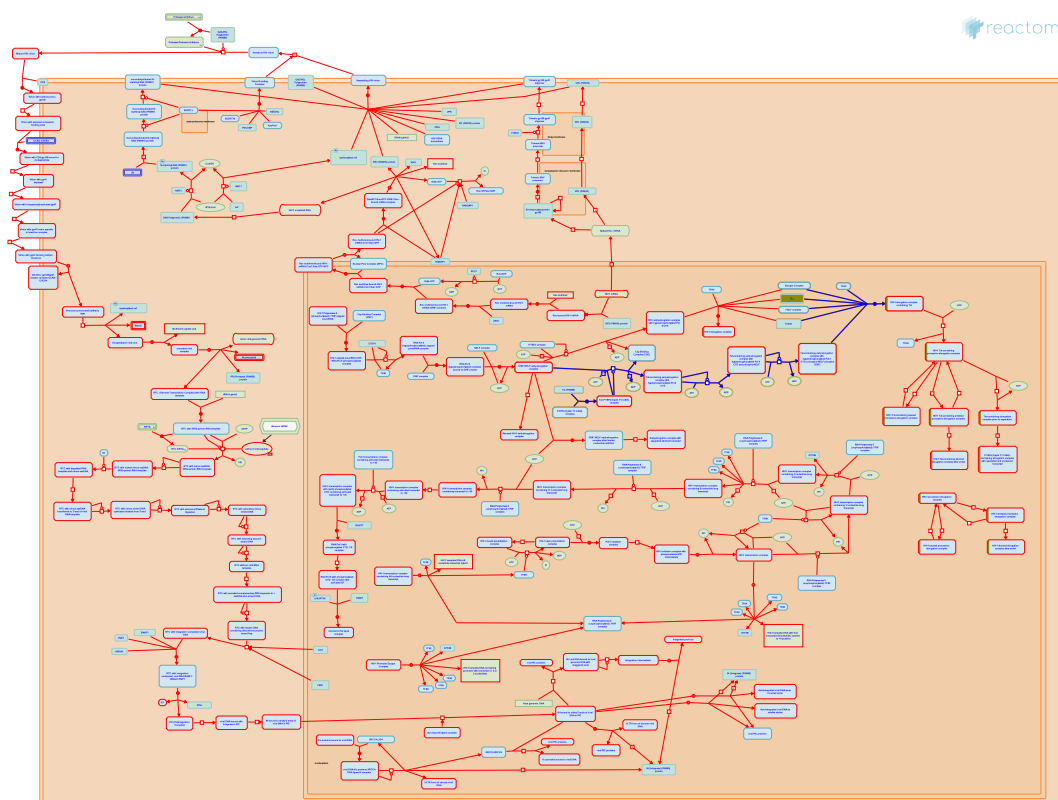
### Edit history

| Date       | Action   | Author     |
|------------|----------|------------|
| 2005-10-14 | Created  | Matthews L |
| 2023-03-08 | Modified | Matthews L |

### 1 submitted entities found in this pathway, mapping to 1 Reactome entities

| Input | UniProt Id |
|-------|------------|
| ELL   | P55199     |

## 9. Formation of HIV-1 elongation complex containing HIV-1 Tat (R-HSA-167200)



**Cellular compartments:** nucleoplasm.

**Diseases:** Human immunodeficiency virus infectious disease.

**Inferred from:** Formation of RNA Pol II elongation complex .

This HIV-1 event was inferred from the corresponding human RNA Pol II transcription event in Reactome. The details relevant to HIV-1 are described below. For a more detailed description of the general mechanism, see the link to the corresponding RNA Pol II transcription event below.

The formation of the HIV-1 elongation complex involves Tat mediated recruitment of P-TEFb(Cyclin T1:Cdk9) to the TAR sequence (Wei et al, 1998) and P-TEFb(Cyclin T1:Cdk9) mediated phosphorylation of the RNA Pol II CTD as well as the negative transcriptional elongation factors DSIF and NELF (Herrmann, 1995; Ivanov et al. 2000; Fujinaga et al. 2004; Zhou et al., 2004).

### References

- Gaynor RB, Kwak YT, Ivanov D & Guo J (2000). Domains in the SPT5 protein that modulate its transcriptional regulatory properties. *Mol Cell Biol*, 20, 2970-83. [🔗](#)
- Rice AP & Herrmann CH (1995). Lentivirus Tat proteins specifically associate with a cellular protein kinase, TAK, that hyperphosphorylates the carboxyl-terminal domain of the large subunit of RNA polymerase II: candidate for a Tat cofactor. *J Virol*, 69, 1612-20. [🔗](#)
- Peterlin BM, Irwin D, Taube R, Huang Y, Fujinaga K & Kurosu T (2004). Dynamics of human immunodeficiency virus transcription: P-TEFb phosphorylates RD and dissociates negative effectors from the transactivation response element. *Mol Cell Biol*, 24, 787-95. [🔗](#)
- Wei P, Jones KA, Fang SM, Fischer WH & Garber ME (1998). A novel CDK9-associated C-type cyclin interacts directly with HIV-1 Tat and mediates its high-affinity, loop-specific binding to TAR RNA. *Cell*, 92, 451-62. [🔗](#)

Kashanchi F, Brady JN, Deng L, Park HU, Zhou M, Pumfery A, ... Kumar A (2004). Coordination of transcription factor phosphorylation and histone methylation by the P-TEFb kinase during human immunodeficiency virus type 1 transcription. J Virol, 78, 13522-33. [🔗](#)

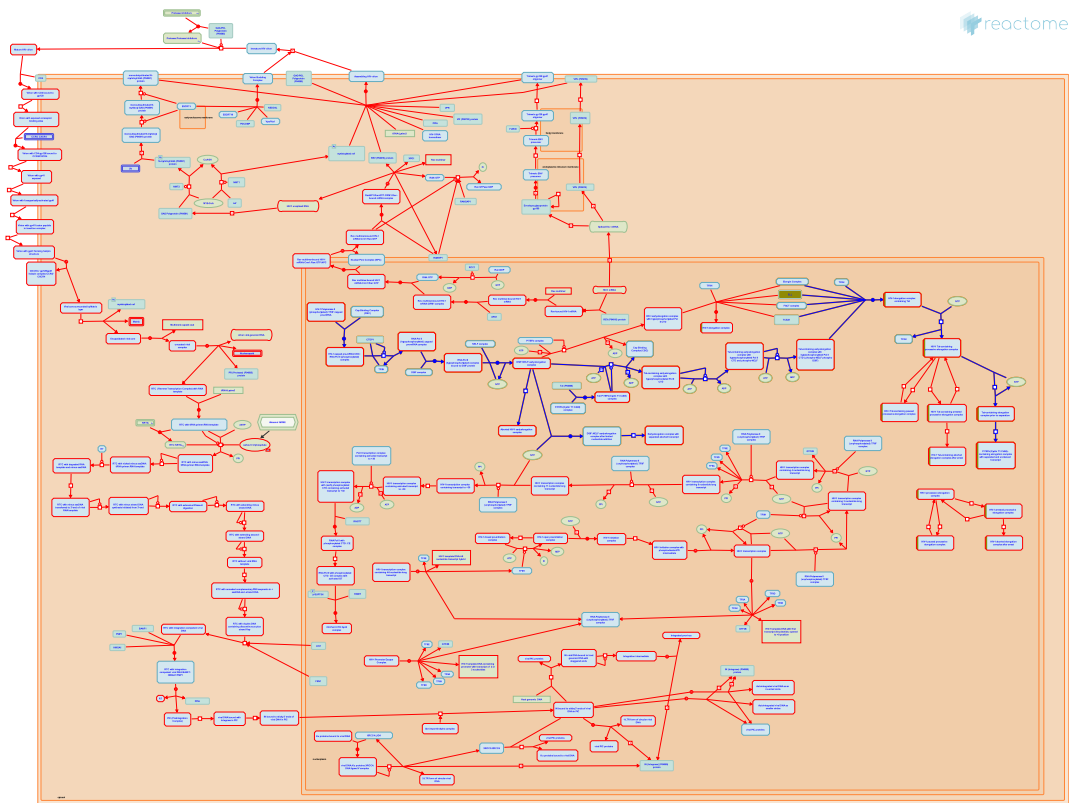
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| 2005-07-27 | Authored | Rice AP, Matthews L |
| 2005-10-12 | Created  | Matthews L          |
| 2023-03-08 | Modified | Matthews L          |

### 1 submitted entities found in this pathway, mapping to 1 Reactome entities

| Input | UniProt Id |
|-------|------------|
| ELL   | P55199     |

# 10. HIV Transcription Elongation (R-HSA-167169)



**Cellular compartments:** nucleoplasm.

**Diseases:** Human immunodeficiency virus infectious disease.

**Inferred from:** RNA Polymerase II Transcription Elongation.

In the absence of the HIV-1 protein Tat, transcription of the proviral DNA is inefficient and results in the production of truncated transcripts (Kao et al., 1987). While initiation of transcription from the HIV-1 LTR and formation of the early elongation complex occurs normally, transcription elongation is incomplete with non-processive polymerases disengaging from the proviral DNA template prematurely (reviewed in Karn 1999). The mechanism of Tat-mediated elongation is described below.

## References

Karn J (1999). Tackling Tat. J Mol Biol, 293, 235-54. [🔗](#)

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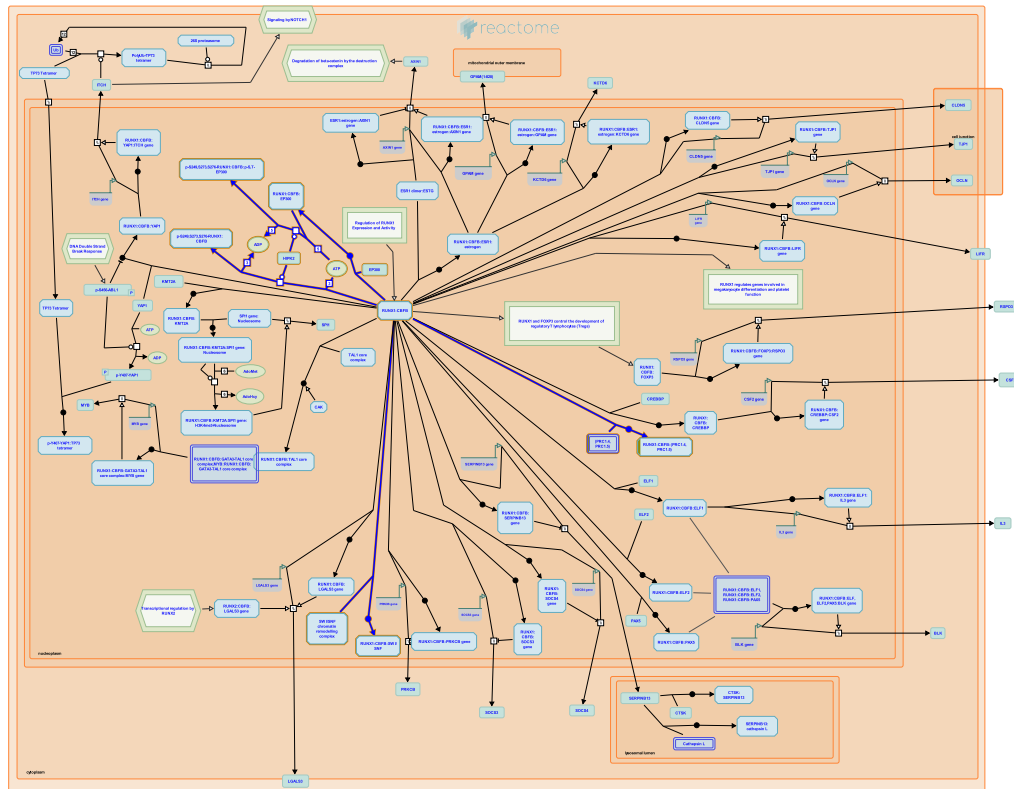
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| 2005-07-27 | Edited   | Matthews L          |
| 2005-07-27 | Authored | Rice AP, Matthews L |
| 2005-10-11 | Created  | Matthews L          |
| 2023-03-08 | Modified | Matthews L          |

**1 submitted entities found in this pathway, mapping to 1 Reactome entities**

| Input | UniProt Id |
|-------|------------|
| ELL   | P55199     |

## 11. RUNX1 interacts with co-factors whose precise effect on RUNX1 targets is not known (R-HSA-8939243)



The transcriptional activity of the RUNX1:CBFB complex is regulated by interaction with co-factors and posttranslational modifications of RUNX1. Protein serine/threonine kinase HIPK2 can phosphorylate RUNX1 and affect transcriptional activity of the RUNX1:CBFB complex during hematopoiesis. Some CBFB mutations found in leukemia interfere with HIPK2-mediated phosphorylation of RUNX1. HIPK2 can simultaneously phosphorylate RUNX1 and EP300 (p300) bound to the RUNX1:CBFB1 complex (Aikawa et al. 2006, Wee et al. 2008).

The RUNX1:CBFB complex can associate with the polycomb repressor complex 1 (PRC1). PRC1 complexes are found at many RUNX1 target promoters and can act either as co-activators or co-repressors in the transactivation of RUNX1 targets (Yu et al. 2011).

RUNX1 recruits the SWI/SNF chromatin remodeling complex to many RUNX1 target promoters by directly interacting with several SWI/SNF subunits (Bakshi et al. 2010).

Other co-factors of the RUNX1:CBFB complex are annotated in the context of transcriptional regulation of specific genes.

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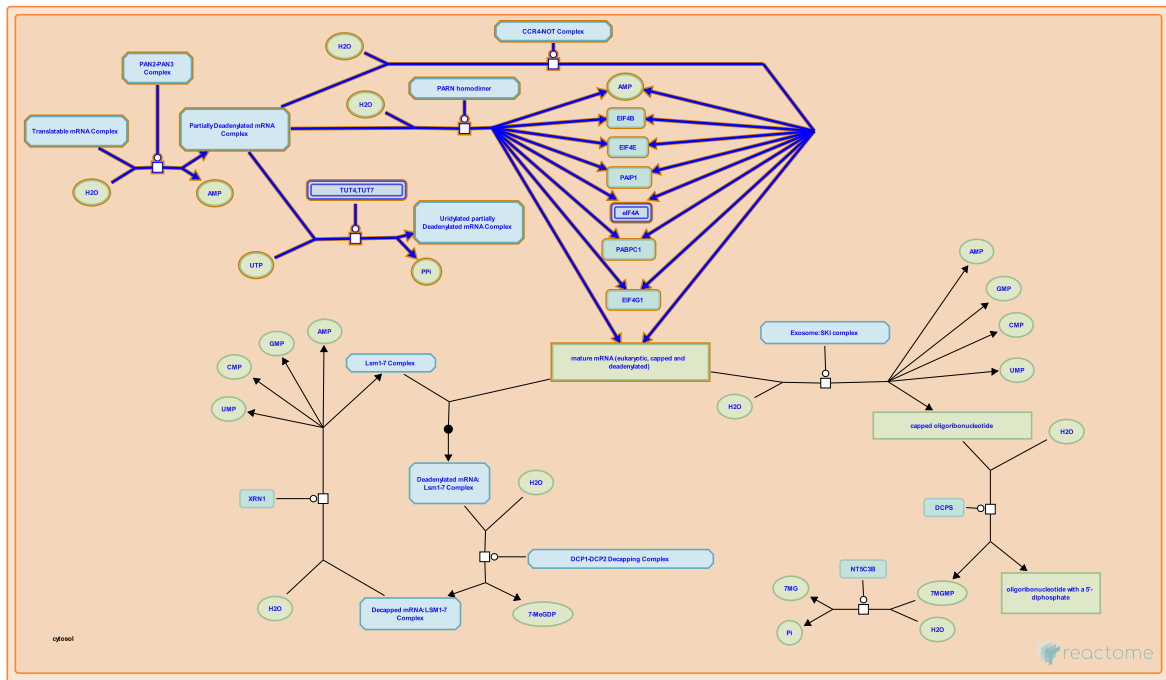
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| 2016-09-14 | Authored | Orlic-Milacic M  |
| 2016-09-16 | Created  | Orlic-Milacic M  |
| 2016-12-20 | Reviewed | Ito Y, Chuang LS |
| 2017-05-09 | Edited   | Orlic-Milacic M  |
| 2023-08-26 | Modified | Wright A         |

### 1 submitted entities found in this pathway, mapping to 1 Reactome entities

| Input | UniProt Id |
|-------|------------|
| CBX6  | O95503     |

## 12. Deadenylation of mRNA (R-HSA-429947)



**Cellular compartments:** cytosol.

Deadenylation of mRNA proceeds in two steps. According to current models, in the first step the poly(A) tail is shortened from about 200 adenosine residues to about 80 residues by the PAN2-PAN3 complex. In the second step the poly(A) tail is further shortened to 10-15 residues by either the CCR4-NOT complex or by the PARN exoribonuclease. How a particular mRNA is targeted to CCR4-NOT or PARN is unknown.

A number of other deadenylase enzymes can be identified in genomic searches. One particularly interesting one is nocturnin, a protein that is related to the CCR-1 deadenylase and plays a role in circadian rhythms.

There is also evidence for networking between deadenylation and other aspects of gene expression. CCR4-NOT, for example, is known to be a transcription factor. PARN is part of a complex that regulates poly(A) tail length and hence translation in developing oocytes.

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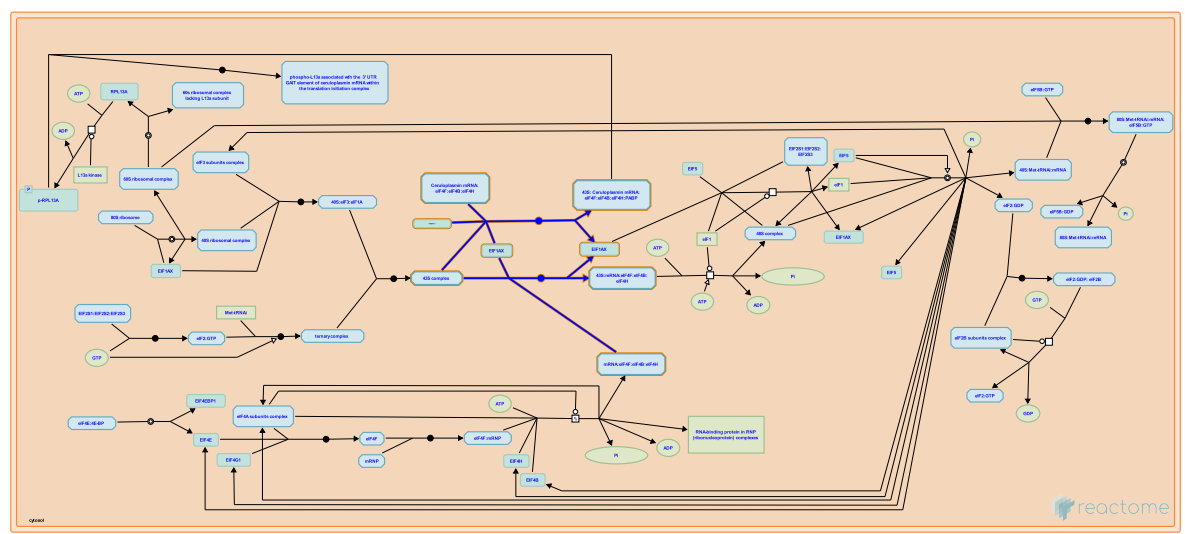
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| 2009-07-22 | Edited   | May B    |
| 2009-07-22 | Authored | May B    |
| 2009-07-24 | Created  | May B    |
| 2009-09-17 | Reviewed | Wilusz J |
| 2023-08-26 | Modified | Wright A |

## Interactors found in this pathway (1)

| Input | UniProt Id | Interacts with | Input | UniProt Id | Interacts with |
|-------|------------|----------------|-------|------------|----------------|
| CBX6  | O95503     | P11940         |       |            |                |

13. Translation initiation complex formation (R-HSA-72649)



Cellular compartments: cytosol.

The translation initiation complex forms when the 43S complex binds the mRNA that is associated with eIF4F, eIF4B and eIF4H. eIF4G in the eIF4F complex can directly contact eIF3 in the 43S complex. eIF1A is necessary for the formation of this complex.

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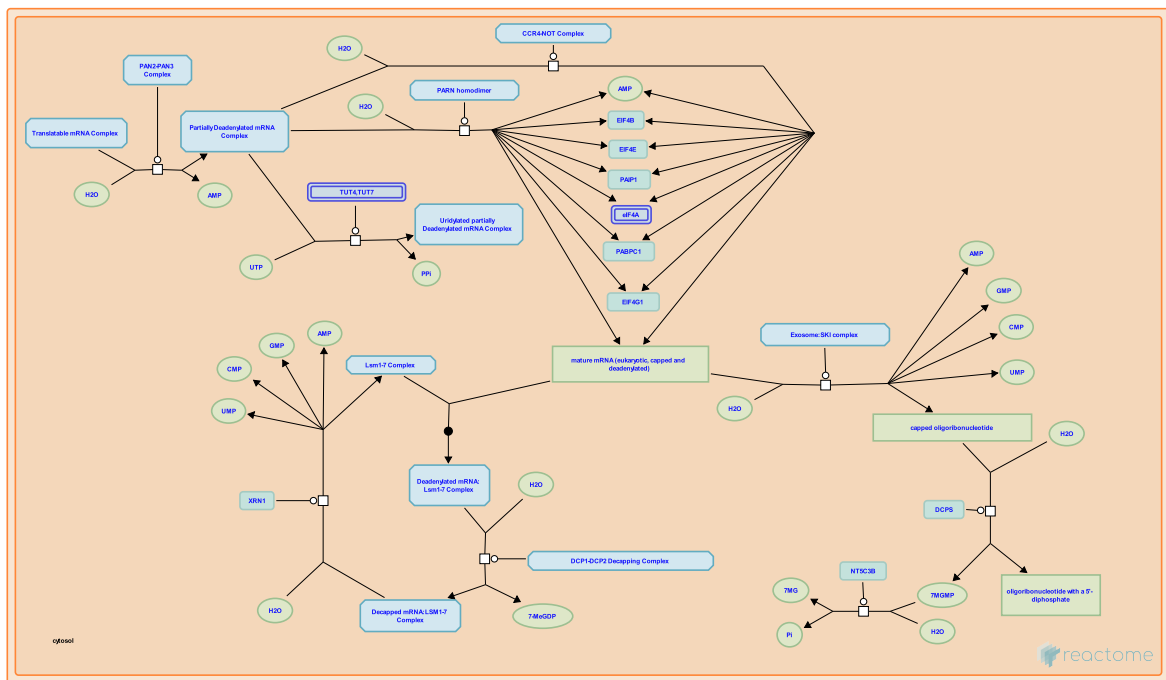
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| 2002-12-16 | Created  | Merrick WC |
| 2023-08-26 | Modified | Wright A   |

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| Input | UniProt Id | Interacts with | Input | UniProt Id | Interacts with |
|-------|------------|----------------|-------|------------|----------------|
| CBX6  | O95503     | P11940         |       |            |                |

#### 14. Deadenylation-dependent mRNA decay ([R-HSA-429914](#))



**Cellular compartments:** cytosol.

After undergoing rounds of translation, mRNA is normally destroyed by the deadenylation-dependent pathway. Though the trigger is unclear, deadenylation likely proceeds in two steps: one catalyzed by the PAN2-PAN3 complex that shortens the poly(A) tail from about 200 adenosine residues to about 80 residues and one catalyzed by the CCR4-NOT complex or by the PARN enzyme that shortens the tail to about 10-15 residues.

After deadenylation the mRNA is then hydrolyzed by either the 5' to 3' pathway or the 3' to 5' pathway. It is unknown what determinants target a mRNA to one pathway or the other.

The 5' to 3' pathway is initiated by binding of the Lsm1-7 complex to the 3' oligoadenylate tail followed by decapping by the DCP1-DCP2 complex. The 5' to 3' exoribonuclease XRN1 then hydrolyzes the remaining RNA.

The 3' to 5' pathway is initiated by the exosome complex at the 3' end of the mRNA. The exosome processively hydrolyzes the mRNA from 3' to 5', leaving only a capped oligoribonucleotide. The cap is then removed by the scavenging decapping enzyme DCPS.

## References

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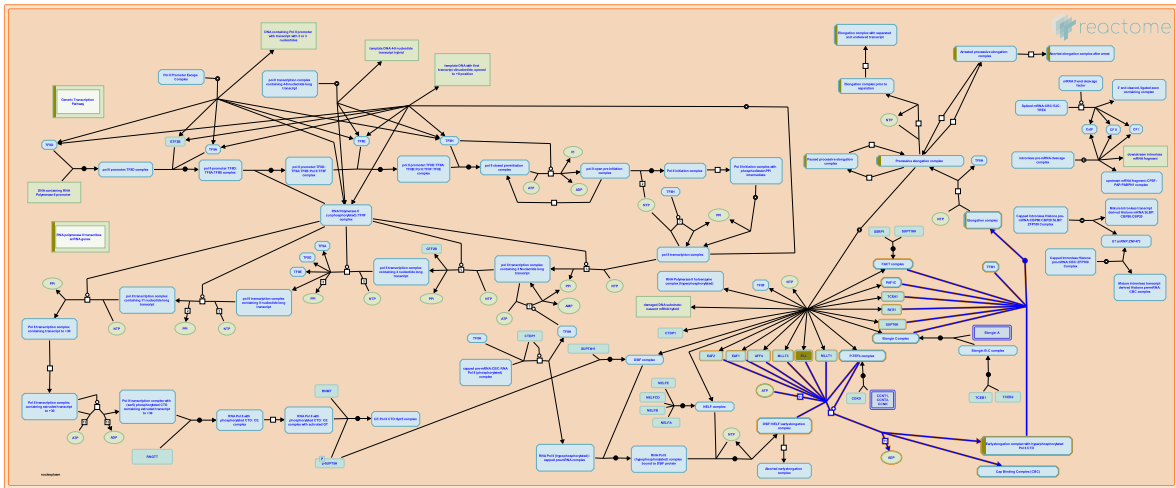
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| 2009-07-22 | Authored | May B    |
| 2009-07-24 | Created  | May B    |
| 2009-09-17 | Reviewed | Wilusz J |
| 2023-08-26 | Modified | Wright A |

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| Input | UniProt Id | Interacts with | Input | UniProt Id | Interacts with |
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| CBX6  | O95503     | P11940         |       |            |                |

15. Formation of RNA Pol II elongation complex (R-HSA-112382)



**Cellular compartments:** nucleoplasm.

TFIIS is a transcription factor involved in different phases of transcription, occurring in a major ubiquitous form and other tissue specific forms. TFIIS stimulates RNA Pol II complex out of elongation arrest.

Other transcription factors like ELL, Elongin family members and TFIIF interact directly with elongating Pol II and increase its elongation rate. These factors have been observed to act on naked DNA templates by suppressing transient pausing by the enzyme at all or most steps of nucleotide addition. In *Drosophila*, ELL is found at a large number of transcriptionally active sites on polytene chromosomes. In general, ELL is suspected to have more unidentified functions.

Elongin is a heterotrimeric protein complex that stimulates the overall rate of elongation. In addition, Elongin may act as an E3 Ubiquitin ligase. Ubiquitylation of RNA Pol II occurs rapidly after genotoxic assault by UV light or chemicals, and results in degradation by proteasome. The FACT complex appears to promote elongation by facilitating passage of polymerase through chromatin.

All these factors contribute to the formation of a processive elongation complex centered around the RNA Pol II complex positioned on the DNA:RNA hybrid. This enables the RNA Pol II elongation complex to function as a platform that coordinates mRNA processing and export (Reviewed by Shilatifard et al., 2003).

**References**

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**Edit history**

| Date       | Action   | Author                 |
|------------|----------|------------------------|
| 2003-09-11 | Created  | Conaway JW, Conaway RC |
| 2004-06-22 | Authored | Gopinathrao G          |
| 2023-08-26 | Modified | Wright A               |

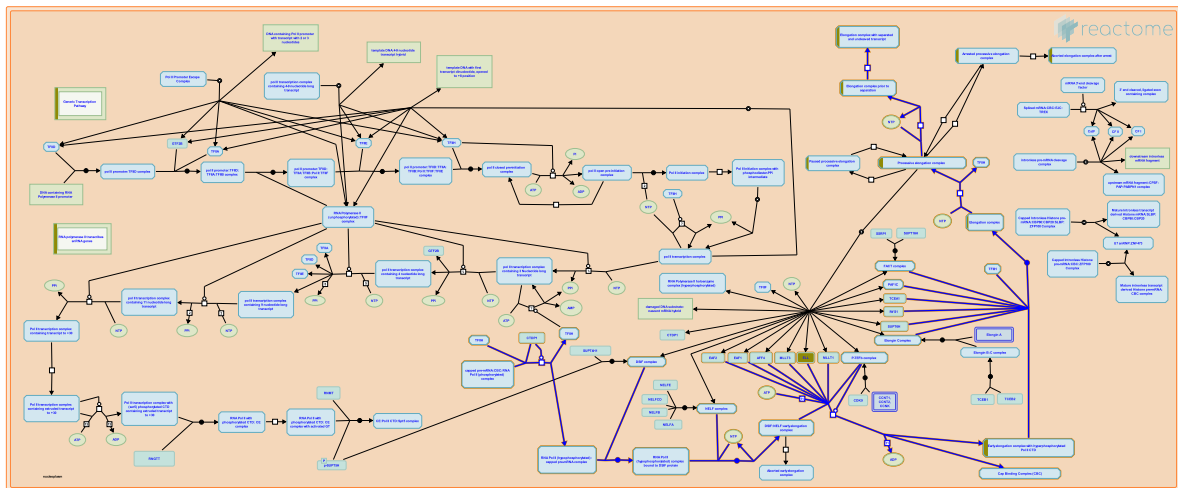
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|-------|------------|----------------|-------|------------|----------------|
| ELL   | P55199     | Q9UHB7, Q96JC9 |       |            |                |

## 16. RNA Polymerase II Transcription Elongation (R-HSA-75955)



**Cellular compartments:** nucleoplasm.

The mechanisms governing the process of elongation during eukaryotic mRNA synthesis are being unraveled by recent studies. These studies have led to the expected discovery of a diverse collection of transcription factors that directly regulate the activities of RNA Polymerase II and unexpected discovery of roles for many elongation factors in other basic processes like DNA repair, recombination, etc. The transcription machinery and structural features of the major RNA polymerases are conserved across species. The genes active during elongation fall under different classes like, housekeeping, cell-cycle regulated, development and differentiation specific genes etc. The list of genes involved in elongation has been growing in recent times, and include: -TFIIS, DSIF, NELF, P-Tefb etc. that are involved in drug induced or sequence-dependent arrest - TFIIF, ELL, elongin, elongator etc. that are involved in increasing the catalytic rate of elongation by altering the  $K_m$  and/or the  $V_{max}$  of Pol II - FACT, Paf1 and other factors that are involved chromatin modification - DNA repair proteins, RNA processing and export factors, the 19S proteasome and a host of other factors like Spt5-Spt5, Paf1, and NELF complexes, FCP1P etc. (Arndt and Kane, 2003). Elongation also represents processive phase of transcription in which the activities of several mRNA processing factors are coupled to transcription through their binding to RNA polymerase (Pol II). One of the key events that enables this interaction is the differential phosphorylation of Pol II CTD. Phosphorylation pattern of CTD changes during transcription, most significantly at the beginning and during elongation process. TFIIF-dependent Ser5 phosphorylation is observed primarily at promoter regions while P-Tefb mediated Ser2 phosphorylation is seen mainly in the coding regions, during elongation. Experimental evidence suggests a dynamic association of RNA processing factors with differently modified forms of the polymerase during the transcription cycle. (Komarnitsky et al., 2000). [Komarnitsky et al 2000, Arndt & Kane 2003, Shilatifard et al 2003]

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## Edit history

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| 2003-09-11 | Authored | Conaway JW, Conaway RC |
| 2003-09-11 | Created  | Conaway JW, Conaway RC |
| 2023-08-24 | Edited   | Joshi-Tope G           |
| 2023-08-26 | Modified | Wright A               |

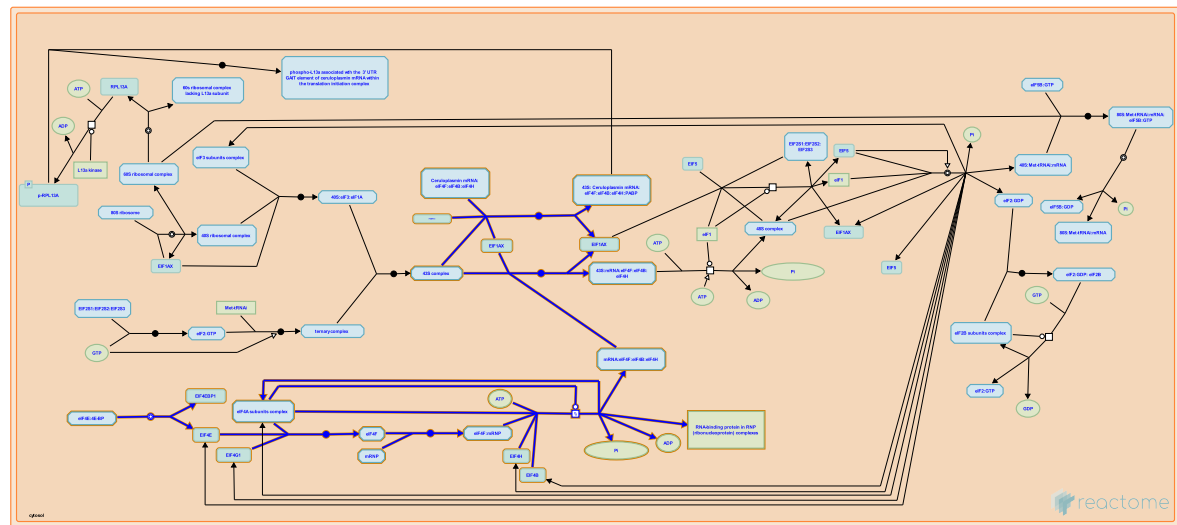
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|-------|------------|----------------|-------|------------|----------------|
| ELL   | P55199     | Q9UHB7, Q96JC9 |       |            |                |

17. Activation of the mRNA upon binding of the cap-binding complex and eIFs, and subsequent binding to 43S (R-HSA-72662)



Cellular compartments: cytosol.

The cap-binding complex is constituted by the initiation factors eIF4A, eIF4G and eIF4E. First, eIF4E must be released from the inactive eIF4E:4E-BP complex. Then eIF4A interacts with eIF4G, and eIF4E binds to the amino-terminal domain of eIF4G, resulting in the formation of the cap-binding complex eIF4F. eIF4A together with eIF4B or eIF4H is thought to unwind RNA secondary structures near the 5'-end of the mRNA. The translation initiation complex is formed when the 43S complex binds the cap-bound mRNA.

References

Tahara SM, Morgan MA, Grifo JA, Merrick WC & Shatkin AJ (1983). New initiation factor activity required for globin mRNA translation. J Biol Chem, 258, 5804-10. [🔗](#)

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Benne R & Hershey JW (1978). The mechanism of action of protein synthesis initiation factors from rabbit reticulocytes. J Biol Chem, 253, 3078-87. [🔗](#)

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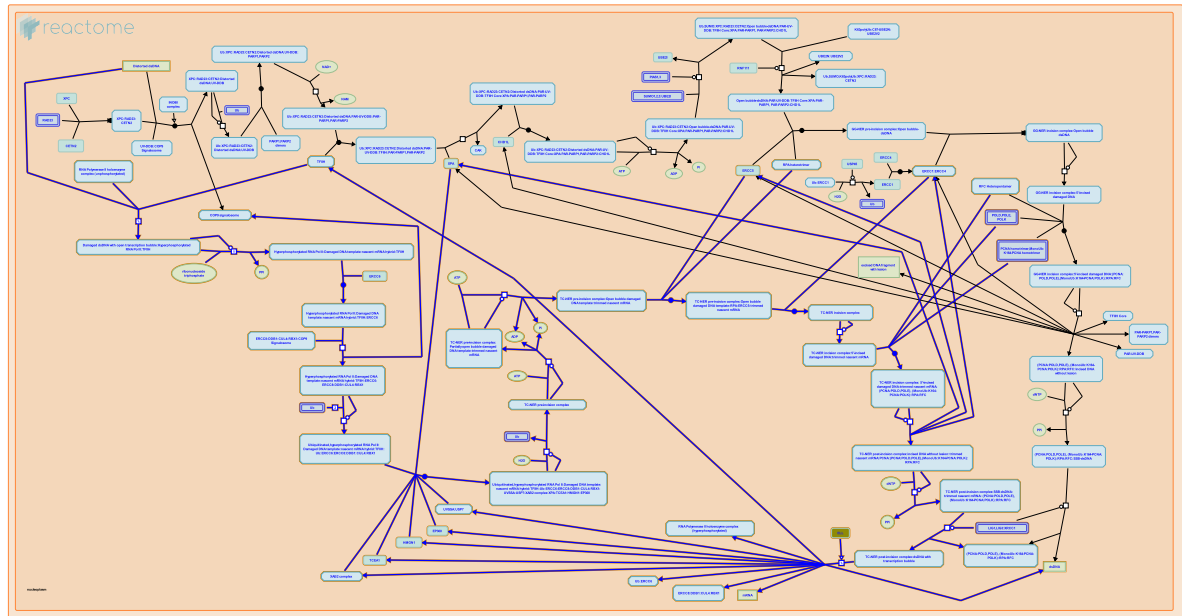
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| 2023-08-26 | Modified | Wright A   |

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| CBX6  | O95503     | P11940         |       |            |                |

## 18. Transcription-Coupled Nucleotide Excision Repair (TC-NER) (R-HSA-6781827)



**Cellular compartments:** nucleoplasm.

DNA damage in transcribed strands of active genes is repaired through a specialized nucleotide excision repair (NER) pathway known as transcription-coupled nucleotide excision repair (TC-NER). TC-NER impairment is the underlying cause of a severe hereditary disorder Cockayne syndrome, an autosomal recessive disease characterized by hypersensitivity to UV light.

TC-NER is triggered by helix distorting lesions that block the progression of elongating RNA polymerase II (RNA Pol II). Stalled RNA Pol II complex triggers the recruitment of ERCC6. ERCC6, commonly known as CSB (Cockayne syndrome protein B) recruits ERCC8, commonly known as CSA (Cockayne syndrome protein A). ERCC8 has 7 WD repeat motifs and is part of the ubiquitin ligase complex that also includes DDB1, CUL4A or CUL4B and RBX1. The ERCC8 ubiquitin ligase complex is one of the key regulators of TC-NER that probably exerts its role by ubiquitinating one or more factors involved in this repair process, including the RNA Pol II complex and ERCC6.

In addition to RNA Pol II, ERCC6 and the ERCC8 complex, the transcription elongation factor TFIIH, which is also involved in global genome nucleotide excision repair (GG-NER), is recruited to sites of TC-NER. The TC-NER pre-incision complex also includes XPA, XAB2 complex, TCEA1 (TFIIS), HMG1, UVSSA in complex with USP7, and EP300 (p300). XPA probably contributes to the assembly and stability of the pre-incision complex, similar to its role in GG-NER. The XAB2 complex is involved in pre-mRNA splicing and may modulate the structure of the nascent mRNA hybrid with template DNA through its RNA-DNA helicase activity, allowing proper processing of DNA damage. TCEA1 may be involved in RNA Pol II backtracking, which allows repair proteins to gain access to the damage site. It also facilitates trimming of the 3' end of protruding nascent mRNA from the stalled RNA Pol II, enabling recovery of RNA synthesis after repair.

Deubiquitinating activity of the UVSSA:USP7 complex is needed for ERCC6 stability at repair sites. Non-histone nucleosomal binding protein HMG1 and histone acetyltransferase p300 (EP300) re-model the chromatin around the damaged site, thus facilitating repair.

Dual incision of the lesion-containing oligonucleotide from the affected DNA strand is performed by two DNA endonucleases, the ERCC1:ERCC4 (ERCC1:XPF) complex and ERCC5 (XPG), which also participate in GG-NER. DNA polymerases delta, epsilon or kappa fill in the single stranded gap after dual incision and the remaining single strand nick is sealed by DNA ligases LIG1 or LIG3 (the latter in complex with XRCC1), similar to GG-NER. After the repair of DNA damage is complete, RNA Pol II resumes RNA synthesis.

For past and recent reviews, see Mellon et al. 1987, Svejstrup 2002, Hanawalt and Spivak 2008, Vermeulen and Fousteri 2013 and Marteijn et al. 2014.

## References

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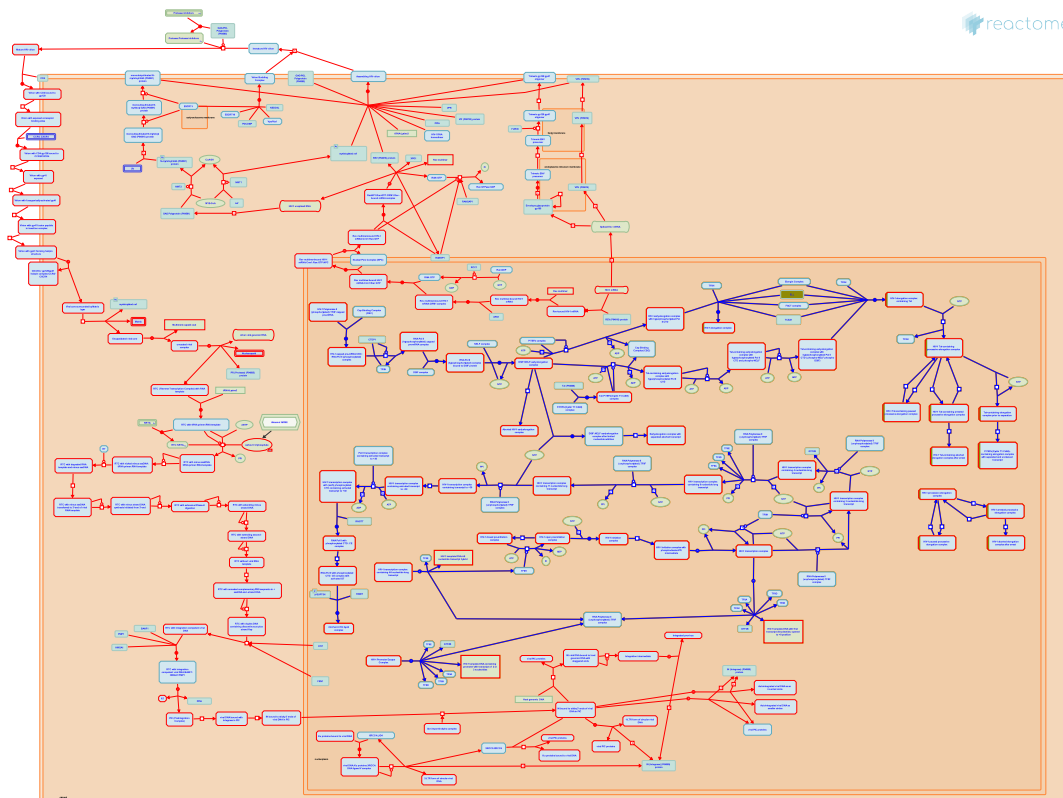
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| 2015-06-02 | Created  | Orlic-Milacic M |
| 2015-06-16 | Revised  | Orlic-Milacic M |
| 2015-06-16 | Edited   | Orlic-Milacic M |
| 2015-06-16 | Authored | Orlic-Milacic M |
| 2015-08-03 | Reviewed | Fousteri M      |
| 2023-08-31 | Modified | Wright A        |

## 1 submitted entities found in this pathway, mapping to 1 Reactome entities

| Input | UniProt Id |
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| ELL   | P55199     |

## 19. Transcription of the HIV genome (R-HSA-167172)



**Cellular compartments:** nucleoplasm.

**Diseases:** Human immunodeficiency virus infectious disease.

**Inferred from:** RNA Polymerase II Transcription.

Expression of the integrated HIV-1 provirus is dependent on the host cell Pol II transcription machinery, but is regulated in critical ways by HIV-1 Tat and Rev proteins. The long terminal repeats (LTR) located at either end of the proviral DNA contain regulatory sequences that recruit cellular transcription factors. The U3 region of the 5' LTR contains numerous cis-acting elements that regulate Pol II-mediated transcription initiation. The full-length transcript, which encodes nine genes, functions as an mRNA and is packaged as genomic RNA. Smaller (subgenomic) viral mRNAs are generated by alternative splicing. The activities of Tat and Rev create two phases of gene expression (see Karn 1999; Cullen 1991). The Tat protein is an RNA specific trans-activator of LTR-mediated transcription. Association of Tat with TAR, a RNA stem-loop within the RNA leader sequence, is required for efficient elongation of the HIV-1 transcript. In the early phase of viral transcription, a multiply-spliced set of mRNAs is generated, producing the transcripts of the regulatory proteins, Tat, Rev, and Nef. In the late phase, Rev regulates nuclear export of HIV-1 mRNAs, repressing expression of the early regulatory mRNAs and promoting expression of viral structural proteins. Nuclear export of the unspliced and partially spliced late HIV-1 transcripts that encode the structural proteins requires the association of Rev with a cis-acting RNA sequence in the transcripts (Rev Response Element, RRE).

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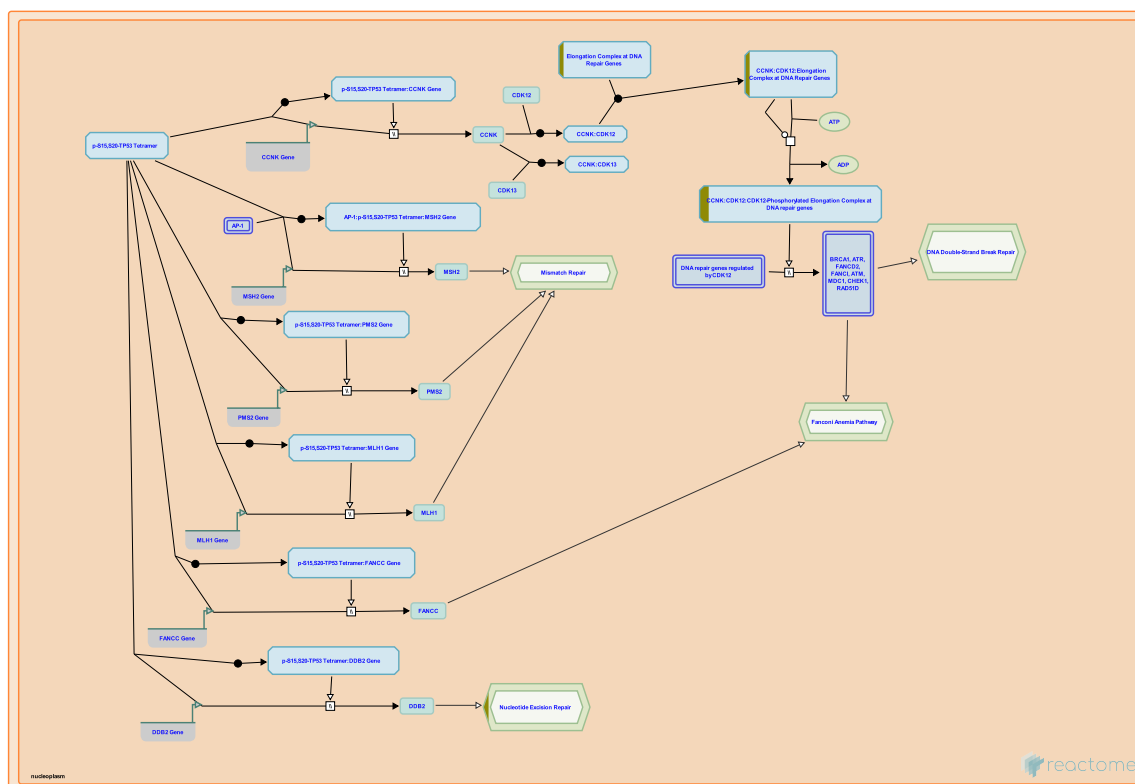
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| 2005-01-05 | Reviewed | Peterlin BM         |
| 2005-07-27 | Edited   | Matthews L          |
| 2005-07-27 | Authored | Rice AP, Matthews L |
| 2005-10-11 | Created  | Matthews L          |
| 2023-03-08 | Modified | Matthews L          |

**1 submitted entities found in this pathway, mapping to 1 Reactome entities**

| Input | UniProt Id |
|-------|------------|
| ELL   | P55199     |

## 20. TP53 Regulates Transcription of DNA Repair Genes ([R-HSA-6796648](#))



Several DNA repair genes contain p53 response elements and their transcription is positively regulated by TP53 (p53). TP53-mediated regulation probably ensures increased protein level of DNA repair genes under genotoxic stress.

TP53 directly stimulates transcription of several genes involved in DNA mismatch repair, including MSH2 (Scherer et al. 2000, Warnick et al. 2001), PMS2 and MLH1 (Chen and Sadowski 2005). TP53 also directly stimulates transcription of DDB2, involved in nucleotide excision repair (Tan and Chu 2002), and FANCC, involved in the Fanconi anemia pathway that repairs DNA interstrand crosslinks (Liebetrau et al. 1997). Other p53 targets that can influence DNA repair functions are RRM2B (Kuo et al. 2012), XPC (Fitch et al. 2003), GADD45A (Amundson et al. 2002), CDKN1A (Cazzalini et al. 2010) and PCNA (Xu and Morris 1999). Interestingly, the responsiveness of some of these DNA repair genes to p53 activation has been shown in human cells but not for orthologous mouse genes (Jegga et al. 2008, Tan and Chu 2002). Contrary to the positive modulation of nucleotide excision repair (NER) and mismatch repair (MMR), p53 can negatively modulate base excision repair (BER), by down-regulating the endonuclease APEX1 (APE1), acting in concert with SP1 (Poletto et al. 2016).

Expression of several DNA repair genes is under indirect TP53 control, through TP53-mediated stimulation of cyclin K (CCNK) expression (Mori et al. 2002). CCNK is the activating cyclin for CDK12 and CDK13 (Blazek et al. 2013). The complex of CCNK and CDK12 binds and phosphorylates the C-terminal domain of the RNA polymerase II subunit POLR2A, which is necessary for efficient transcription of long DNA repair genes, including BRCA1, ATR, FANCD2, FANCI, ATM, MDC1, CHEK1 and RAD51D. Genes whose transcription is regulated by the complex of CCNK and CDK12 are mainly involved in the repair of DNA double strand breaks and/or the Fanconi anemia pathway (Blazek et al. 2011, Cheng et al. 2012, Bosken et al. 2014, Bartkowiak and Greenleaf 2015, Ekumi et al. 2015).

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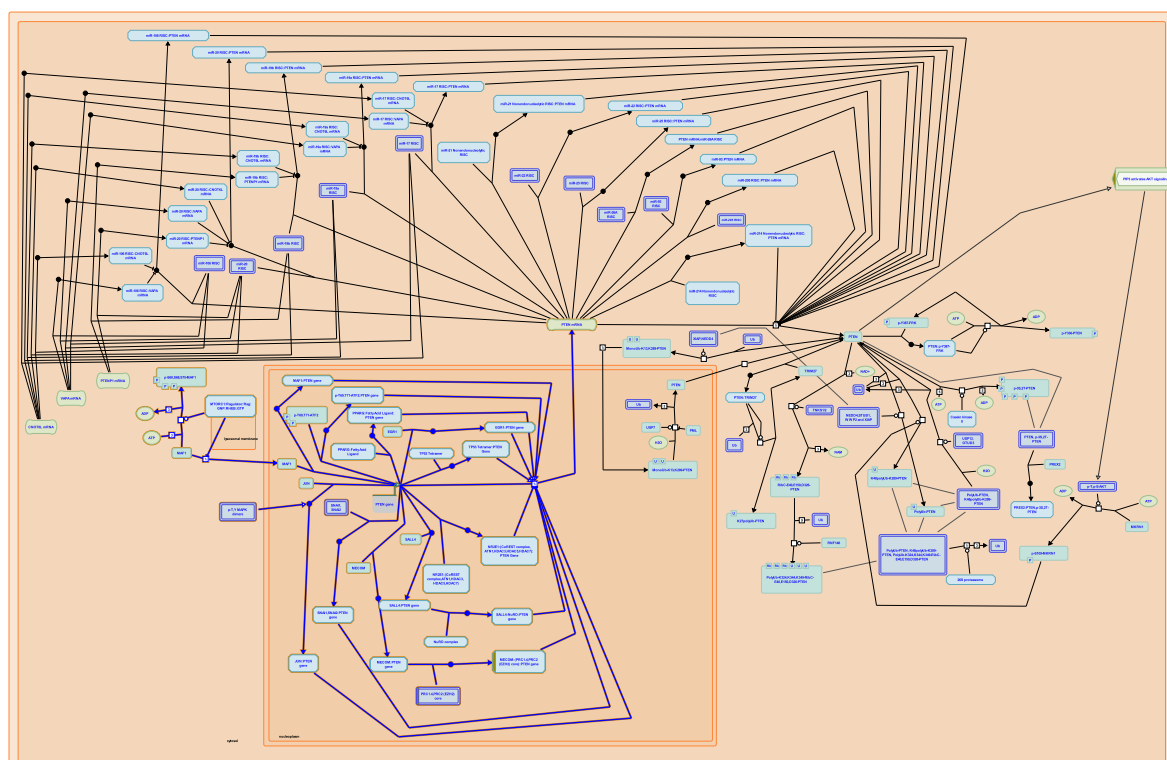
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| Date       | Action   | Author            |
|------------|----------|-------------------|
| 2015-09-05 | Created  | Orlic-Milacic M   |
| 2015-10-14 | Edited   | Orlic-Milacic M   |
| 2015-10-14 | Authored | Orlic-Milacic M   |
| 2016-02-04 | Reviewed | Zaccara S, Inga A |
| 2023-08-31 | Modified | Wright A          |

## 1 submitted entities found in this pathway, mapping to 1 Reactome entities

| Input | UniProt Id |
|-------|------------|
| ELL   | P55199     |

## 21. Regulation of PTEN gene transcription (R-HSA-8943724)



reactome

Transcription of the PTEN gene is regulated at multiple levels. Epigenetic repression involves the recruitment of Mi-2/NuRD upon SALL4 binding to the PTEN promoter (Yang et al. 2008, Lu et al. 2009) or EVI1-mediated recruitment of the polycomb repressor complex (PRC) to the PTEN promoter (Song et al. 2009, Yoshimi et al. 2011). Transcriptional regulation is also elicited by negative regulators, including NR2E1:ATN1 (atrophin-1) complex, JUN (c-Jun), SNAIL and SLUG (Zhang et al. 2006, Vasudevan et al. 2007, Escriva et al. 2008, Uygur et al. 2015) and positive regulators such as TP53 (p53), MAF1, ATF2, EGR1 or PPARG (Stambolic et al. 2001, Virolle et al. 2001, Patel et al. 2001, Shen et al. 2006, Li et al. 2016).

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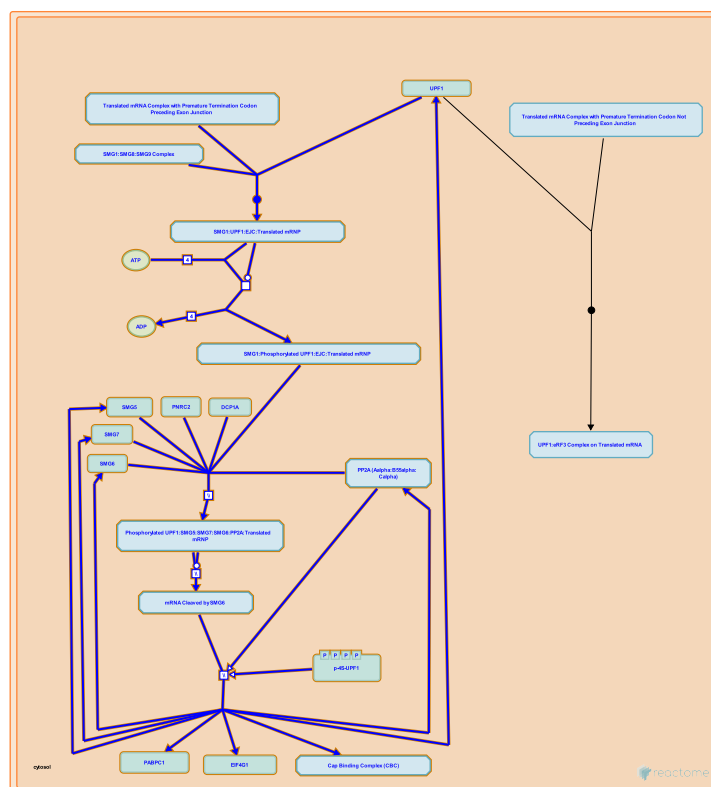
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| 2015-10-29 | Authored | Orlic-Milacic M        |
| 2016-08-11 | Authored | Salmena L, Carracedo A |
| 2016-10-28 | Created  | Orlic-Milacic M        |
| 2017-05-09 | Edited   | Orlic-Milacic M        |
| 2023-08-26 | Modified | Wright A               |

**1 submitted entities found in this pathway, mapping to 1 Reactome entities**

| Input | UniProt Id |
|-------|------------|
| CBX6  | O95503     |

## 22. Nonsense Mediated Decay (NMD) enhanced by the Exon Junction Complex (EJC) (R-HSA-975957)



**Cellular compartments:** cytosol.

During normal translation termination eRF3 associates with the ribosome and then interacts with PABP bound to the polyadenylate tail of the mRNA to release the ribosome and allow a new round of translation to commence. Nonsense-mediated decay (NMD) is triggered if eRF3 at the ribosome interacts with UPF1, which may compete with PABP (reviewed in Isken and Maquat 2007, Chang et al. 2007, Behm-Ansmant et al. 2007, Rebbapragada and Lykke-Andersen 2009, Bhuvanagiri et al. 2010, Nicholson et al. 2010, Durand and Lykke-Andersen 2011). An exon junction located 50-55 nt downstream of a termination codon is observed to enhance NMD.

Exon-junction complexes (EJCs) are deposited on the mRNA during splicing in the nucleus, remain on mRNAs after transport to the cytosol, and are dislodged by the ribosome as it progresses along the mRNA during the pioneer round of translation (Gehring et al. 2009). EJCs contain the core factors eIF4A-III, Magoh-Y14, and CASC3 as well as the peripheral factors RNPS1, UPF2, and UPF3. UPF2 and UPF3 recruit UPF1 to eRF3 at the terminating ribosome. Thus an EJC downstream of a termination codon will not have been dislodged during translation and will recruit UPF1, triggering NMD.

UPF1 is believed to form a complex containing SMG1, SMG8, and SMG9. In the key regulatory step of NMD SMG1 phosphorylates UPF1. The phosphorylated UPF1 then recruits either SMG6 or SMG5 and SMG7. SMG6 is itself an endoribonuclease that cleaves the mRNA. SMG5 and SMG7 do not have endoribonuclease activity, but are thought to recruit ribonucleases. Nonsense-mediated decay has been observed to involve deadenylation, decapping, and both 5' to 3' and 3' to 5' exonuclease activities, but the exact degradative pathways taken by a given mRNA are not yet known.

UPF1 also plays roles in Staufen-mediated decay, histone mRNA decay, telomere maintenance, genome integrity, and may play a role in normal termination of translation.

## References

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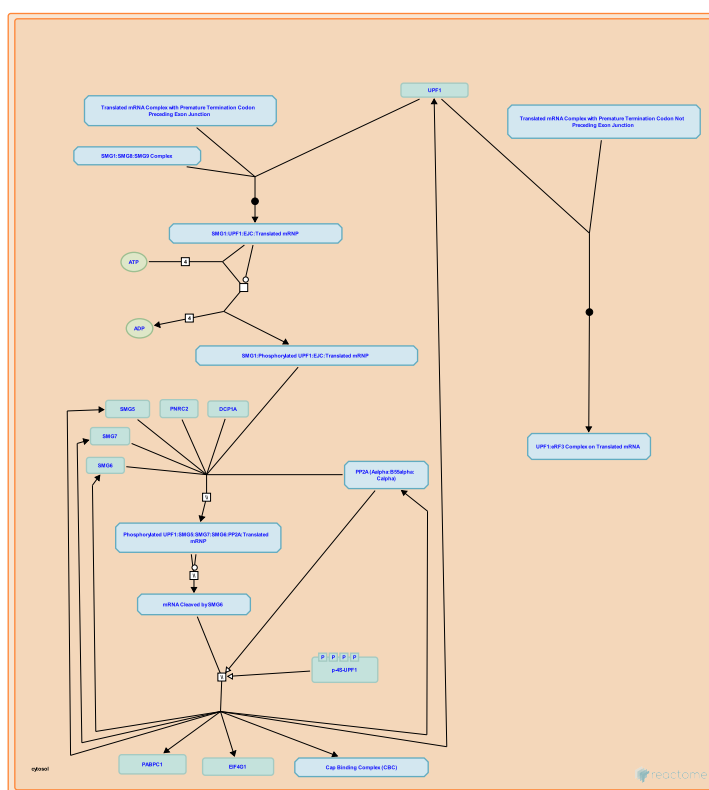
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| 2010-10-08 | Edited   | May B       |
| 2010-10-08 | Authored | May B       |
| 2010-10-11 | Created  | May B       |
| 2011-05-19 | Reviewed | Neu-Yilik G |
| 2023-08-31 | Modified | Wright A    |

## Interactors found in this pathway (1)

| Input | UniProt Id | Interacts with | Input | UniProt Id | Interacts with |
|-------|------------|----------------|-------|------------|----------------|
| CBX6  | O95503     | P11940         |       |            |                |

## 23. Nonsense-Mediated Decay (NMD) (R-HSA-927802)



**Cellular compartments:** cytosol.

The Nonsense-Mediated Decay (NMD) pathway activates the destruction of mRNAs containing premature termination codons (PTCs) (reviewed in Isken and Maquat 2007, Chang et al. 2007, Behm-Ansmant et al. 2007, Neu-Yilik and Kulozik 2008, Rebbapragada and Lykke-Andersen 2009, Bhuvanagiri et al. 2010, Nicholson et al. 2010, Durand and Lykke-Andersen 2011). In mammalian cells a termination codon can be recognized as premature if it precedes an exon-exon junction by at least 50-55 nucleotides or if it is followed by an abnormal 3' untranslated region (UTR). While length of the UTR may play a part, the qualifications for being "abnormal" have not been fully elucidated. Also, some termination codons preceding exon junctions are not degraded by NMD so the criteria for triggering NMD are not yet fully known (reviewed in Rebbapragada and Lykke-Andersen 2009). While about 30% of disease-associated mutations in humans activate NMD, about 10% of normal human transcripts are also degraded by NMD (reviewed in Stalder and Muhlemann 2008, Neu-Yilik and Kulozik 2008, Bhuvanagiri et al. 2010, Nicholson et al. 2010). Thus NMD is a normal physiological process controlling mRNA stability in unmutated cells.

Exon junction complexes (EJCs) are deposited on an mRNA during splicing in the nucleus and are displaced by ribosomes during the first round of translation. When a ribosome terminates translation the A site encounters the termination codon and the eRF1 factor enters the empty A site and recruits eRF3. Normally, eRF1 cleaves the translated polypeptide from the tRNA in the P site and eRF3 interacts with Polyadenylate-binding protein (PABP) bound to the polyadenylated tail of the mRNA.

During activation of NMD eRF3 interacts with UPF1 which is contained in a complex with SMG1, SMG8, and SMG9. NMD can arbitrarily be divided into EJC-enhanced and EJC-independent pathways. In EJC-enhanced NMD, an exon junction is located downstream of the PTC and the EJC remains on the mRNA after termination of the pioneer round of translation. The core EJC is associated with UPF2 and UPF3, which interact with UPF1 and stimulate NMD. Once bound near the PTC, UPF1 is phosphorylated by SMG1. The phosphorylation is the rate-limiting step in NMD and causes UPF1 to recruit either SMG6, which is an endoribonuclease, or SMG5 and SMG7, which recruit ribonucleases. SMG6 and SMG5:SMG7 recruit phosphatase PP2A to dephosphorylate UPF1 and allow further rounds of degradation. How EJC-independent NMD is activated remains enigmatic but may involve competition between PABP and UPF1 for eRF3.

## References

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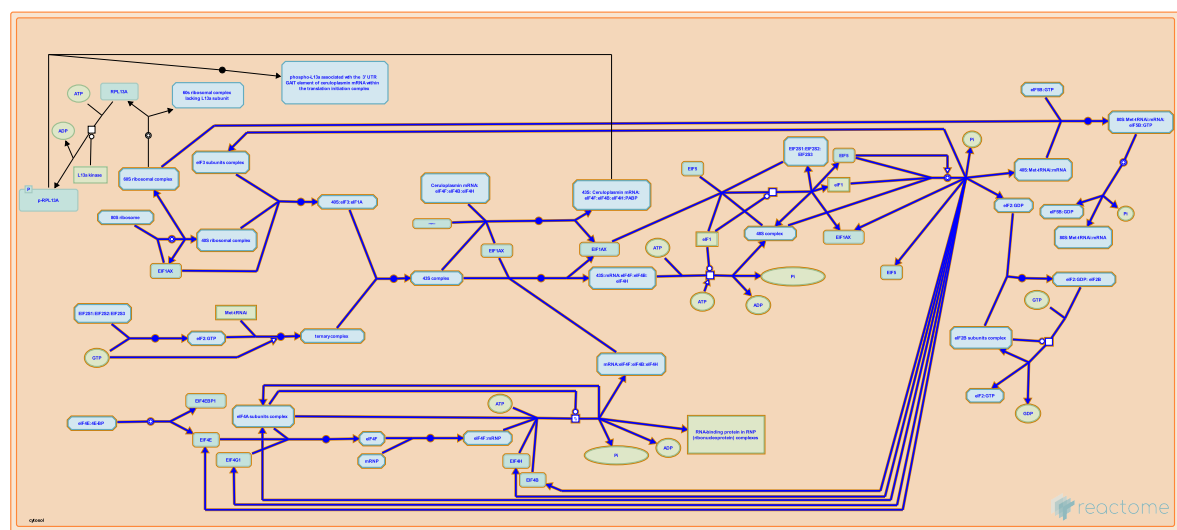
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| 2010-08-06 | Edited   | May B       |
| 2010-08-06 | Authored | May B       |
| 2010-08-10 | Created  | May B       |
| 2011-05-19 | Reviewed | Neu-Yilik G |
| 2023-08-26 | Modified | Wright A    |

## Interactors found in this pathway (1)

| Input | UniProt Id | Interacts with | Input | UniProt Id | Interacts with |
|-------|------------|----------------|-------|------------|----------------|
| CBX6  | O95503     | P11940         |       |            |                |

24. Cap-dependent Translation Initiation (R-HSA-72737)



Translation initiation is a complex process in which the Met-tRNA<sub>i</sub> initiator, 40S, and 60S ribosomal subunits are assembled by eukaryotic initiation factors (eIFs) into an 80S ribosome at the start codon of an mRNA. The basic mechanism for this process can be described as a series of five steps: 1) formation of a pool of free 40S subunits, 2) formation of the ternary complex (Met-tRNA<sub>i</sub>/eIF2/GTP), and subsequently, the 43S complex (comprising the 40S subunit, Met-tRNA<sub>i</sub>/eIF2/GTP, eIF3 and eIF1A), 3) activation of the mRNA upon binding of the cap-binding complex eIF4F, and factors eIF4A, eIF4B and eIF4H, with subsequent binding to the 43S complex, 4) ribosomal scanning and start codon recognition, and 5) GTP hydrolysis and joining of the 60S ribosomal subunit.

References

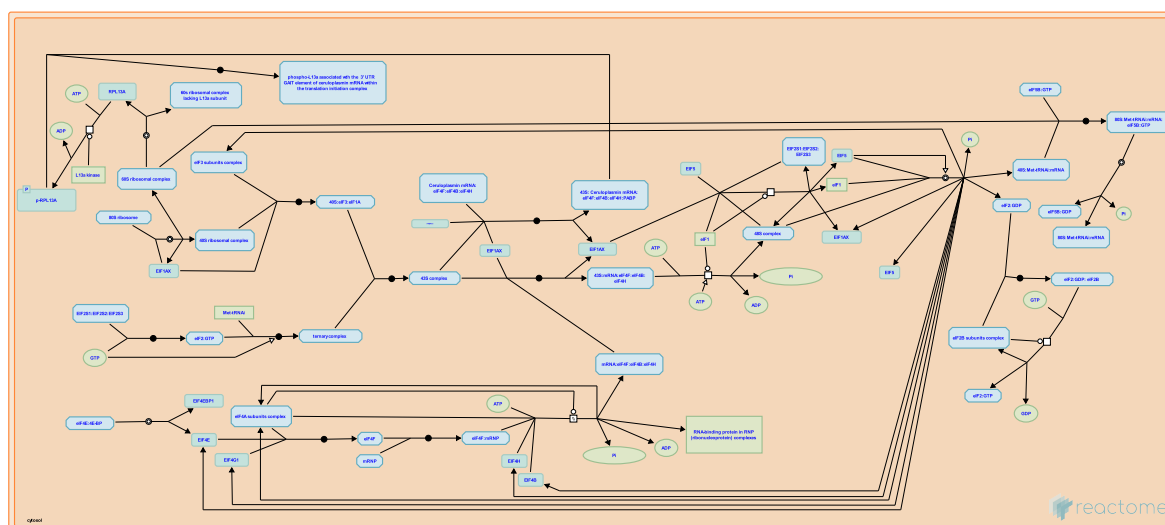
Edit history

| Date       | Action   | Author     |
|------------|----------|------------|
| 2002-12-16 | Created  | Merrick WC |
| 2023-08-26 | Modified | Wright A   |

Interactors found in this pathway (1)

| Input | UniProt Id | Interacts with | Input | UniProt Id | Interacts with |
|-------|------------|----------------|-------|------------|----------------|
| CBX6  | O95503     | P11940         |       |            |                |

## 25. Eukaryotic Translation Initiation (R-HSA-72613)



Initiation of translation in the majority of eukaryotic cellular mRNAs depends on the 5'-cap (m7GpppN) and involves ribosomal scanning of the 5' untranslated region (5'-UTR) for an initiating AUG start codon. Therefore, this mechanism is often called cap-dependent translation initiation. Proximity to the cap, as well as the nucleotides surrounding an AUG codon, influence the efficiency of the start site recognition during the scanning process. However, if the recognition site is poor enough, scanning ribosomal subunits will ignore and skip potential starting AUGs, a phenomenon called leaky scanning. Leaky scanning allows a single mRNA to encode several proteins that differ in their amino-termini. Merrick (2010) provides an overview of this process and highlights several features of it that remain incompletely understood.

Several eukaryotic cell and viral mRNAs initiate translation by an alternative mechanism that involves internal initiation rather than ribosomal scanning. These mRNAs contain complex nucleotide sequences, called internal ribosomal entry sites, where ribosomes bind in a cap-independent manner and start translation at the closest downstream AUG codon.

Initiation on several viral and cellular mRNAs is cap-independent and is mediated by binding of the ribosome to internal ribosome entry site (IRES) elements. These elements are often found in characteristically long structured regions on the 5'-UTR of an mRNA that may or may not have regulatory upstream open reading frames (uORFs). Both of these features on the 5'-end of the mRNA hinder ribosomal scanning, and thus promote a cap-independent translation initiation mechanism. IRESs act as specific translational enhancers that allow translation initiation to occur in response to specific stimuli and under the control of different trans-acting factors, as for example when cap-dependent protein synthesis is shut off during viral infection. Such regulatory elements have been identified in the mRNAs of growth factors, protooncogenes, angiogenesis factors, and apoptosis regulators, which are translated under a variety of stress conditions, including hypoxia, serum deprivation, irradiation and apoptosis. Thus, cap-independent translational control might have evolved to regulate cellular responses in acute but transient stress conditions that would otherwise lead to cell death, while the same mechanism is of major importance for viral mRNAs to bypass the shutting-off of host protein synthesis after infection. Encephalomyocarditis virus (EMCV) and hepatitis C virus exemplify two distinct mechanisms of IRES-mediated initiation. In contrast to cap-dependent initiation, the eIF4A and eIF4G subunits of eIF4F bind immediately upstream of the EMCV initiation codon and promote binding of a 43S complex. Accordingly, EMCV initiation does not involve scanning and does not require eIF1, eIF1A, and the eIF4E subunit of eIF4F. Nonetheless, initiation on some EMCV-like IRESs requires additional non-canonical initiation factors, which alter IRES conformation and promote binding of eIF4A/eIF4G. Initiation on the hepatitis C virus IRES is simpler: a 43S complex containing only eIF2 and eIF3 binds directly to the initiation codon as a result of specific interaction of the IRES and the 40S subunit.

## References

Merrick WC (2010). Eukaryotic protein synthesis: still a mystery. *J Biol Chem*, 285, 21197-201. [🔗](#)

## Edit history

| Date       | Action   | Author     |
|------------|----------|------------|
| 2002-12-16 | Created  | Merrick WC |
| 2023-08-26 | Modified | Wright A   |

## Interactors found in this pathway (1)

| Input | UniProt Id | Interacts with | Input | UniProt Id | Interacts with |
|-------|------------|----------------|-------|------------|----------------|
| CBX6  | O95503     | P11940         |       |            |                |

## 6. Identifiers found

Below is a list of the input identifiers that have been found or mapped to an equivalent element in Reactome, classified by resource.

**2 of the submitted entities were found, mapping to 3 Reactome entities**

| Input | UniProt Id | Input | UniProt Id     |
|-------|------------|-------|----------------|
| CBX6  | O95503     | ELL   | P55199, Q9HB65 |

### Interactors (3)

| Input | UniProt Id | Interacts with | Input | UniProt Id | Interacts with |
|-------|------------|----------------|-------|------------|----------------|
| CBX6  | O95503     | P11940         | ELL   | P55199     | Q9UHB7, Q96JC9 |
| WDR37 | Q9Y2I8     | Q96ES7         |       |            |                |